

Goldbach Deserts

Construction, Optimisation, and the Limits of Covering

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Abstract

For an even integer N , let $\mathfrak{g}(N)$ denote the least prime occurring in a Goldbach partition of N . Empirically $\mathfrak{g}$ is minuscule—it never exceeds 9781 for any even $N \leq 4 \cdot 10^{18}$ —and the interesting question is therefore extremal and constructive: how large can $\mathfrak{g}(N)$ be *forced* to be, and what must be paid in the size of N ? This thesis studies that question along three record games: the unbounded *height* game, the *threshold* game $T(Q)$ (minimise N subject to $\mathfrak{g}(N) > Q$), and the *budget* game $R(X)$ (maximise $\mathfrak{g}(N)$ subject to $N < X$).

Chapters 2 and 3 develop the constructive theory. A congruence cover on the prime offsets, realised through the Chinese Remainder Theorem and completed by a progression search, produces certified integers far smaller than the classical primorial or complete-cover constructions. We give a cost model for such searches—a single *failure exponent* $E = |U| \cdot \text{boost} / \log N$ whose exponential e^E predicts the number of candidates that must be examined—and validate it against 1.2×10^9 scanned candidates, where the measured and predicted exponents agree to within 0.1 nats. The programme’s certified incumbents are a 2 480-digit integer with $\mathfrak{g}(N) = 1\,157\,341$, a 150-digit integer with $\mathfrak{g}(N) = 104\,527$, and a 199-digit integer with $\mathfrak{g}(N) = 119\,419$.

Chapters 4–6 ask where this stops, and this is where the thesis makes its main contributions. We show that the difficulty of the whole enterprise is an *integrality gap*: the natural linear-programming relaxation of the covering problem is *vacuous*—fractionally, the available moduli cover 100% of the prime offsets—so the entire failure exponent is the price of rounding a fractional cover to an integral one. We then attack that gap with exact and global optimisation (large-neighbourhood search with certified windows, belief propagation with decimation, banded constructions) and find it immovable at the scales tested, with evidence that the greedy construction already sits at or near the quenched optimum.

Finally we prove, at the level of rigour of a well-supported research programme rather than a finished theorem, a *closure* result: within a concrete language of certificate schemas, no construction beats covering. Each family closes for a different and quantifiable reason—a value-set ceiling of $\sqrt{Q}$, a supply-limited purchase plan that runs $2.5\times$ over the available design entropy, a cluster bound, a lattice-isolation argument, or an empty intersection at record scale. The theorem is conditional on two named hypotheses, one of which—the absence of a compositeness test cheaper than a single modular exponentiation—we isolate and pose as an open problem of independent interest.

The closure theorem was stress-tested by an adversarial search (an LLM-driven evolutionary system) against a machine-checkable scorer. Its first success was not against the theorem but against our *referee*: it found a $50\times$ accounting error in the evaluator and exploited it before any claim was published. The audit protocol caught it, and repairing the referee strengthened the mathematics. We record this as a methodological finding: adversarial search against a formal claim is most valuable as a referee-hardening device.

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Chapter 1

Introduction: three games and one exponent

Goldbach's conjecture asserts that every even integer $N > 2$ is a sum of two primes; it has been verified for all $N \leq 4 \cdot 10^{18}$ [1]. Heuristically one expects not just one representation but many. The Hardy–Littlewood circle method—which estimates such counts by treating the primes as a random set of the appropriate density and then correcting for divisibility constraints—predicts [3]

$$r(N) \sim 2\Pi_2 \frac{N}{(\log N)^2} \prod_{\substack{p|N \\ p>2}} \frac{p-1}{p-2}, \quad \Pi_2 = \prod_{p>2} \left(1 - \frac{1}{(p-1)^2}\right), \quad (1.1)$$

where $r(N) = \#\{p \leq N : p \text{ and } N-p \text{ both prime}\}$. Reading this from left to right: $N/(\log N)^2$ is the naive guess, since a number near N is prime with probability about $1/\log N$ and there are N choices for p ; the twin-prime constant $\Pi_2 \approx 0.6601$ corrects for the fact that primes are not truly random modulo small numbers; and the last factor inflates the count when N is highly composite. The conclusion is that a typical large even integer should have *many* Goldbach partitions. So the interesting question is not whether a representation exists, but how lopsided the most lopsided one must be.

Definition 1.1. For an even integer $N \geq 4$ admitting a Goldbach representation, the *least Goldbach summand* is

$$\mathfrak{g}(N) = \min\{p \text{ prime} : N-p \text{ is prime}\}.$$

This is the quantity $p(n)$ of the *minimal Goldbach partition* studied by Granville, van de Lune and te Riele [4], who conjectured a ceiling

$$\mathfrak{g}(N) = O((\log N)^2 \log \log N). \quad (1.2)$$

Empirically $\mathfrak{g}$ is tiny and grows very slowly; its record values are catalogued in the OEIS [5] (A020481, with records A025018/A025019). Exhaustive search gives a slowly climbing ladder: $\mathfrak{g} = 1093$ at $N = 60\,119\,912$; $\mathfrak{g}(N) \leq 5569$ for all even $N \leq 4 \cdot 10^{14}$ [2]; and the current exhaustive record $\mathfrak{g} = 9781$, first attained at $N = 3\,325\,581\,707\,333\,960\,528$, with $\mathfrak{g}(N) \leq 9781$ for *all* even $N \leq 4 \cdot 10^{18}$ [1]. Strikingly, these records occur at integers of at most 19 digits.

We ask the opposite, extremal and constructive question:

How large can $\mathfrak{g}(N)$ be forced to be, and at what cost in the size of N ?

An even integer N for which $\mathfrak{g}(N)$ is large is a *Goldbach desert*: a number all of whose small prime offsets are blocked. Constructing deserts is the subject of this thesis, and the reason the subject is interesting is that every construction turns out to be governed by a single number—an exponent that measures how much luck the construction still needs after all its cleverness is spent.

1.1 The three games

There are three natural record questions, and they reward different constructions. The first is the unbounded *height game*: exhibit as large a value of $\mathfrak{g}(N)$ as possible, with no limit on the size of N . An ordinary prime gap is one way to play, but it is a strictly stronger construction: a gap must block *every* integer offset, while a desert needs to block only the *prime* offsets. Prime-gap constructions are therefore a strict subclass of the constructions available here—a point we return to in §2.4 with a concrete two-orders-of-magnitude example.

For comparisons in which size matters we use two finite games.

Definition 1.2 (Threshold and budget games). For $Q \geq 2$ and $X \geq 4$, define

$$T(Q) = \min\{N \text{ even, Goldbach-representable} : \mathfrak{g}(N) > Q\}, \quad R(X) = \max\{\mathfrak{g}(N) : N < X\}.$$

The *threshold game* fixes Q and tries to lower $T(Q)$; the *budget game* fixes X and tries to raise $R(X)$. “Fewer than 200 decimal digits” means the strict budget $N < 10^{199}$.

Remark 1.3 (A naming caution). The budget game has been written both as $R(10^{199})$ (“fewer than 200 digits”) and as $R(10^{200})$ (“at most 200 digits”) in the literature this thesis builds on, and the two are genuinely different: our 200-digit incumbent with $\mathfrak{g} = 112\,249$ satisfies $N < 10^{200}$ but exceeds 10^{199} by 3.4%, which is precisely why a 199-digit integer with the smaller value $\mathfrak{g} = 119\,419$ is the record for the stricter budget. Throughout we state the bound explicitly with each record.

1.2 Certified incumbents

Table 1.1 lists the programme’s certified incumbents at the time of writing. Every entry has been verified twice: once by the pipeline that produced it and once by an independent verifier written against the record file rather than against the search code, including a re-derivation of every per-offset witness and an independent check of the elliptic-curve primality certificate.

Table 1.1: Certified incumbents. “Witnesses” counts prime offsets excluded by an explicit congruence divisor plus those excluded by a strong base-2 compositeness witness; in every case the positive side is an exhibited prime complement with an ECPP certificate.

Game	Digits of N	$\mathfrak{g}(N)$	Construction
Height (unbounded)	2 480	1 157 341	cover at $Q = 1.16 \cdot 10^6$, $t = 14\,544$
Height, gap-dominating	2 692	1 134 871	cover at $Q = 1\,113\,138$, $k = 14$
Threshold $T(100\,000)$	150	104 527	cover at $Q = 100\,003$, $k = 6.21 \cdot 10^{10}$
Budget $R(10^{199})$	199	119 419	cover at $Q \approx 1.19 \cdot 10^5$, $t = 1.46 \cdot 10^8$
Budget $R(10^{200})$	200	112 249	cover at $Q = 107\,720$, $k = 56\,770$

Two of these deserve comment. The 2692-digit entry is included because it settles the comparison with ordinary prime gaps in the strongest possible way: its $\mathbf{g}$ exceeds the largest *certified* prime gap known (1 113 106), and it does so with an integer roughly 6.9 times shorter than the one realising that gap. Deserts really are cheaper than gaps, by a measurable factor.

The 150-digit threshold record is the endpoint of a *ratchet*: the same covering machinery, re-tuned and re-run, produced a descending ladder $195 \rightarrow 179 \rightarrow 150$ digits at $Q \approx 10^5$, banking a certified record at each rung. Chapter 3 argues that this ratchet discipline, rather than any single clever cover, is the strategy that actually produces records under a fixed compute budget.

1.3 The exponent that governs everything

The technical spine of the thesis is a single quantity. A partial congruence cover fixes N inside an arithmetic progression $N = N_0 + kM$ and disposes of most prime offsets for *every* k ; a residual set U of offsets survives and must be knocked out by luck, one candidate k at a time. Under the standard heuristic, the probability that a given candidate succeeds is e^{-E} with

$$E = \frac{|U| \cdot \text{boost}}{\log N}, \quad \text{boost} = 2 \prod_{r \in \mathcal{R}} \frac{r}{r-1}, \quad (1.3)$$

so that e^E candidates must be examined per expected hit. We call E the *failure exponent*. It is the currency of the entire subject: every design decision—which moduli to use, which residues, how many offsets to leave residual, what digit budget to spend—is a trade in units of E , and every piece of engineering (faster tests, better sieving, selective testing, more hardware) buys a fixed number of nats of E and no more.

Equation (1.3) is a heuristic, so we test it rather than assume it. Across three independent covers and 1.2×10^9 scanned candidates, the measured exponent $\hat{E}$ agrees with the predicted E to within 0.1 nats (Chapter 3). That agreement is what licenses the rest of the thesis: once the model is trusted, questions about feasibility become arithmetic.

1.4 What is new here

Chapters 2 and 3 are largely expository and engineering: they set out the classical constructions, the covering formulation, the calibrated cost model, and the search engine, and they report the certified records and—equally important—the campaigns that failed. The contributions begin in Chapter 4.

1. **The difficulty is an integrality gap** (§4.3). The natural linear-programming relaxation of the covering problem is *vacuous*: fractionally, the available moduli cover 100.00% of the prime offsets at every target we tested. Consequently the whole failure exponent is the price of *rounding*: naive independent rounding leaves 15.6% of offsets uncovered, adaptive greedy leaves 7.56%, and the fractional optimum leaves none. Every absolute percentage point recovered is worth about 2.7 nats, i.e. a factor of 15 in compute.
2. **The rounding gap resists optimisation** (§4.4). We attack it with exact large-neighbourhood search (certified optimal by LP duality on 21 of 22 windows), with belief propagation and decimation, and with a simultaneous exact solve of half the cover at once. The greedy solution $|U| = 867$ is never beaten. A random-restart probe shows why: the landscape is a

field of mutually distant, locally-exact-stable basins, and the adaptive greedy construction lands about 100 offsets deeper than a typical one.

3. **Sub-100-digit deserts reduce to a hard problem** (Chapter 5). Oversized covers would make sub-100-digit deserts easy if their Chinese-remainder representatives could be made small; that requirement is exactly a modular subset-sum problem with per-position choices at density ≈ 1.026 , the regime where neither lattice reduction nor birthday methods work. We also verify against the source that the relevant CRT list-decoding radius is a factor 5.2 short of what the construction would need, and confirm experimentally that lattice methods do not penetrate the Johnson-type radius even when solutions are abundant.
4. **A closure theorem** (Chapter 6). Within a concrete language of certificate schemas, no construction beats covering. Every family closes for its own quantifiable reason—a $\sqrt{Q}$ value-set ceiling, a purchase plan running $2.5\times$ over the design-entropy budget, a cluster bound, lattice isolation, or an empty intersection at record scale—and the theorem is conditional on two named hypotheses, one of which we pose as an open problem.
5. **A methodological finding** (§6.7). We stress-tested the closure theorem with an adversarial evolutionary search against a machine-checkable scorer. Its first success was against the *referee*, not the theorem: it found a $50\times$ accounting error in our evaluator and exploited it within ten candidates. The audit protocol caught it before any claim was made, and repairing the referee strengthened the mathematics.

1.5 Notation and conventions

Throughout, $\log_k$ is the k -fold iterated logarithm (so $\log_2 N = \log \log N$), and each extra log grows *extremely* slowly; $\theta(x) = \sum_{p \leq x} \log p$ is Chebyshev’s function; $\gamma \approx 0.5772$ is Euler’s constant; $f \gg g$ means $f \geq cg$ for some $c > 0$ and all large arguments; $\pi(x)$ is the prime-counting function; and “BPSW” denotes the Baillie–PSW probable-prime test [14, 15]. All entropies, costs and exponents are measured in *nats* (natural logarithm units), because the natural unit of a congruence modulo r is $\log r$ and the natural unit of a search is $\log(\text{candidates})$.

We are careful to separate three claims about any proposed record, and the reader should hold us to the distinction throughout:

- (a) no prime below the stated value is a Goldbach summand of N ;
- (b) N admits a Goldbach representation at all;
- (c) the least summand *equals* the stated value.

A congruence or a failed probable-prime test settles (a) unconditionally—a prime can never fail such a test, so negative results are never in doubt. Claim (b) needs an exhibited prime complement, and (c) needs both together with a primality certificate for that complement. Where a statement is heuristic, conditional, or merely measured, we say so: results proved at the level of rigour of this thesis are marked as such, statements resting on standard but unproved analytic heuristics are flagged, and purely empirical claims are reported with the size of the experiment that supports them.

Chapter 2

Foundations: factorials, primorials, covers

This chapter develops the constructive theory from the ground up. The progression is one of increasing efficiency: a factorial spends each small prime on a single residue class and buys deserts of size $\log N / \log \log N$; a primorial spends each small prime on exactly one offset and buys $\log N$; a congruence cover reuses each prime across a whole residue class and buys $(\log N) \log_2 N \log_4 N / \log_3 N$ unconditionally, which is the best known. Along the way we prove unboundedness of $\mathfrak{g}$ unconditionally, and we settle the relationship with ordinary prime gaps.

2.1 Factorial neighbourhoods and Wilson obstructions

Lemma 2.1 (Factorial composites). *For $n \geq 2$ and $2 \leq k \leq n$, the number $n! + k$ is composite. For $n \geq 4$ and $2 \leq k \leq n$, the number $n! - k$ is also composite.*

Proof. In both cases $k \mid n!$, so $k \mid (n! \pm k)$. For the plus side $n! + k > k$, so $n! + k$ is a proper multiple of k . For the minus side, $n! - k = k(n!/k - 1)$ with $n!/k - 1 \geq 2$ exactly when $n! \geq 3k$, which holds for $n \geq 4$; it fails only at $n = 3, k = 3$, where $3! - 3 = 3$ is prime. $\square$

Lemma 2.2 (Wilson obstructions). *Let $n \geq 2$. (A) If $n + 1$ is prime, then $(n + 1) \mid n! + 1$. (B) If $n + 2$ is prime, then $(n + 2) \mid n! - 1$.*

Recall *Wilson's theorem*: $p > 1$ is prime if and only if $(p - 1)! \equiv -1 \pmod{p}$. We use only the forward direction.

Proof. For (A) put $p = n + 1$; then $n! = (p - 1)! \equiv -1 \pmod{p}$. For (B) put $p = n + 2$; then $(p - 1)! = (n + 1)n!$ and $n + 1 \equiv -1 \pmod{p}$, so $-n! \equiv -1$, i.e. $n! \equiv 1 \pmod{p}$. $\square$

Corollary 2.3. *If $n! + 1 > n + 1$ is prime, then $n + 1$ is composite. In particular the factorial primes $n! + 1$ with $n \geq 3$ (for example $n = 3, 11, 27, 37, 41, \dots$) occur only when $n + 1$ is composite.*

2.1.1 The worked example $N = 27! + 28$

Take $N = 27! + 28$, even, with 29 digits. Since $27! \equiv 0 \pmod{r}$ for every prime $r \leq 27$,

$$N - q = 27! + (28 - q) \equiv 28 - q \pmod{r} \quad (r \leq 27 \text{ prime}), \quad (2.1)$$

so $N - q$ is killed by a small prime r exactly when $r \mid (28 - q)$: a shifted sieve eliminating the class $q \equiv 28 \pmod{r}$.

Proposition 2.4. *For $N = 27! + 28$ every prime q with $2 \leq q \leq 67$ has $N - q$ composite, while $N - 71$ is prime; hence $\mathfrak{g}(N) = 71$.*

Proof by explicit witnesses. Table 2.1 gives, for each prime $q \leq 71$, a factor of $N - q$. For $q \leq 23$ the witness divides $28 - q$ via (2.1). Two cases below the winner are notable: $q = 29$ is killed by 29 because $29 = 27 + 2$ is prime, so Lemma 2.2(B) gives $29 \mid 27! - 1 = N - 29$; and $q = 59$ survives the sieve (as $59 - 28 = 31$ is 27-rough) yet $41 \mid 27! - 31 = N - 59$. Finally $71 - 28 = 43$ is 27-rough and $27! - 43$ is prime. $\square$

Table 2.1: Why every small prime fails as a summand of $N = 27! + 28$.

q	$N - q$	factor	q	$N - q$	factor
2	$27! + 26$	2	31	$27! - 3$	3
3	$27! + 25$	5	37	$27! - 9$	3
5	$27! + 23$	23	41	$27! - 13$	13
7	$27! + 21$	3	43	$27! - 15$	3
11	$27! + 17$	17	47	$27! - 19$	19
13	$27! + 15$	3	53	$27! - 25$	5
17	$27! + 11$	11	59	$27! - 31$	41 (generic)
19	$27! + 9$	3	61	$27! - 33$	3
23	$27! + 5$	5	67	$27! - 39$	3
29	$27! - 1$	29 (Wilson)	71	$27! - 43$	prime

By (2.1) a prime q can even be *tested* only if $q - 28$ is 27-rough, so the viable candidates 29, 59, 71, 89, $\dots$ are sparse and 71 is merely the first with a prime complement. Heuristically the first viable offsets in this family lie on the scale of n , i.e. roughly $\log N / \log \log N$. The factorial is wasteful: it spends the prime r on the single class $q \equiv 28 \pmod{r}$, most of whose members are composite anyway and so did not need killing.

2.2 Primorial deserts, made unconditional

Lemma 2.5 (Primorial core). *Let P be prime, let $m_P = \prod_{p \leq P} p$ be the primorial, and let N be a multiple of m_P with $N > 2P$. Then $N - q$ is composite for every prime $q \leq P$.*

Proof. For a prime $q \leq P$ we have $q \mid m_P \mid N$, hence $q \mid N - q$, and $N - q > P \geq q$, so $N - q$ is a proper multiple of q . $\square$

Lemma 2.5 gives only claim (a) of §1. Adding Dirichlet's theorem upgrades it to an actual Goldbach number, and hence to an unconditional unboundedness result.

Theorem 2.6 (Unconditional unboundedness). *For every prime P put $s = \text{nextprime}(P)$. There are infinitely many primes $t \equiv -s \pmod{m_P}$, and for any such t the even integer $N = s + t$ satisfies $m_P \mid N$, has the explicit partition $N = s + t$, and has $\mathfrak{g}(N) = \text{nextprime}(P) > P$. Hence $\sup_N \mathfrak{g}(N) = \infty$ unconditionally. Moreover the least such t satisfies $\log N \ll P$ by Linnik's theorem [8], so $\mathfrak{g}(N) \gg \log N$.*

Proof. As $\gcd(s, m_P) = 1$, Dirichlet [7] gives infinitely many primes $t \equiv -s \pmod{m_P}$. Then $N = s + t \equiv 0 \pmod{m_P}$ is even. By Lemma 2.5 no prime $\leq P$ is a summand; the only primes below s are those $\leq P$; and $N - s = t$ is prime, so $\mathbf{g}(N) = s$. Linnik bounds the least such prime by $t \ll m_P^L$ for an absolute L , and $\log m_P = \theta(P) \sim P$, so $\log N \ll P \sim \mathbf{g}(N)$. $\square$

Remark 2.7 (Concrete instances). For $P = 13$: $m_{13} = 30030$, $s = 17$, and the least prime $t \equiv -17 \pmod{30030}$ is $t = 30013$, giving $30030 = 17 + 30013$ with $\mathbf{g} = 17$, all primality deterministic. A larger fully proven example is $m_{43} = 13082761331670030$ itself, whose minimal partition is $m_{43} = 89 + 13082761331669941$ (both proven prime, being $< 2^{64}$), so $\mathbf{g}(m_{43}) = 89$ unconditionally.

Remark 2.8 (The roughness bonus). For $N = m_P$, a prime $q > P$ makes $N - q$ coprime to all primes $\leq P$ —*P-rough*—which improves its chance of being prime, since the small primes that catch most composites are already excluded. By Mertens’ theorem ($\prod_{p \leq P} (1 - 1/p) \sim e^{-\gamma} / \log P$) this multiplies the heuristic probability by $\sim e^{\gamma} \log P$, and the expected least surviving summand lands near $P(1 + e^{-\gamma}) \approx 1.56P$. This “boost” factor is not a curiosity: it reappears in (1.3) as the exact multiplier that makes residual offsets harder to kill, and it is the reason a bigger cover is not automatically a better cover (Chapter 4). Either way the primorial pays a modulus $m_P \approx e^P$; the next section removes that waste.

2.3 Prime-offset congruence covers

A primorial spends one prime per summand; a congruence cover reuses each prime on a whole residue class. The one structural constraint is parity: a sum of two odd primes is even, so N is kept even, the prime 2 kills only $q = 2$, and every odd prime below the target must be covered by odd sieving primes.

The strategy is classical. Covering systems were introduced by Erdős [10] precisely to obstruct an additive representation—his infinite progression of odd integers not of the form $2^k + p$ is the direct ancestor of this section, and of the Sierpiński–Riesel tradition that grew from it [34]. What changes here is only the target: the offsets to be blocked are the primes below Q rather than the powers of two.

Proposition 2.9 (Covering certificate). *Let $Q \geq 3$ and let $\{(r, a_r) : r \in \mathcal{R}\}$ be a system of odd primes such that every odd prime $q < Q$ satisfies $q \equiv a_r \pmod{r}$ for some $r \in \mathcal{R}$. If N is even, $N \equiv a_r \pmod{r}$ for all r , and $N > Q + \max_r r$, then $N - q$ is composite for every prime $q < Q$.*

Proof. For $q = 2$, $N - 2$ is even and > 2 . For an odd prime $q < Q$ pick r with $q \equiv a_r \equiv N \pmod{r}$; then $r \mid N - q$ and $N - q > r$, so the divisor is proper. $\square$

Remark 2.10. The size hypothesis only has to make each divisor proper, for which $N > Q + \max_r r$ suffices; the much stronger $N > Q \prod_r r$ is unnecessary. A common solution exists modulo $M = 2 \prod_r r$ by the Chinese Remainder Theorem, and any large enough even representative qualifies.

The covering is built greedily and deterministically.

Algorithm (greedy covering of the odd primes below Q).

1. Let $T = \{\text{odd primes } < Q\}$ and require $N \equiv 0 \pmod{2}$.
2. For $r = 3, 5, 7, 11, \dots$ in increasing order, while $T \neq \emptyset$: let a_r be a residue mod r containing the most elements of T (smallest such residue on ties), require $N \equiv a_r \pmod{r}$, and delete from T every $q \equiv a_r \pmod{r}$.
3. Recover N from the congruences by the Chinese Remainder Theorem.

The resulting modulus is $M = 2 \prod_{r \in \mathcal{R}} r$. For our targets it is far smaller than the primorial $\prod_{p < Q} p$, because each sieving prime is reused across an entire residue class rather than assigned to one offset. Concretely, at $Q = 10^5$ a complete greedy cover uses 356 congruences and a 1 020-digit modulus, against a primorial of some 43 000 digits.

2.3.1 Partial covers and progression search

A complete cover is convenient to verify but is not the cheapest way to make N small. Near the end of the greedy algorithm a new sieving prime may remove only one or two remaining offsets while still multiplying the modulus by r : an expensive way to buy a single compositeness proof. The alternative is to stop early. Let U be the odd prime offsets below Q left uncovered. CRT gives one residue N_0 modulo $M = 2 \prod_{r \in \mathcal{R}} r$, so every candidate has the form

$$N = N_0 + kM, \tag{2.2}$$

and for each $q \in U$ we test $N - q$ directly. A failed probable-prime test is a rigorous compositeness result: pseudoprimes create false positives, never false negatives. So a fast one-sided test is safe as a search filter, while the final surviving complement still needs a certificate.

Proposition 2.11 (Partial-cover certificate). *Let U be the odd primes $q < Q$ not covered by the congruences of Proposition 2.9, and let N be a sufficiently large even solution of those congruences. If $N - q$ is composite for every $q \in U$, then no prime $q < Q$ is a Goldbach summand of N .*

Proof. The congruences prove compositeness for every covered odd prime; parity handles $q = 2$; the hypothesis handles U . $\square$

This is the construction used for every record in this thesis. Its whole character is that it converts a proof obligation into a *search*: the cover is a static prefilter, and the residual set is validated dynamically. How long that search takes is the subject of Chapter 3, and the answer is e^E candidates with E as in (1.3).

2.3.2 Realising a covering unconditionally

Proposition 2.9 still gives only claim (a). The Dirichlet idea of §2.2, applied twice, upgrades any covering to an actual Goldbach number.

Theorem 2.12 (Unconditional realisation). *Let $\{(r, a_r) : r \in \mathcal{R}\}$ cover the odd primes below Q , all r odd primes. Then there are infinitely many even $N > Q + \max_r r$ admitting an explicit partition $N = s + t$ with $N \equiv a_r \pmod{r}$ for all r , and every such N has $\mathfrak{g}(N) \geq Q$.*

Proof. For each r choose $b_r \in \{1, \dots, r-1\} \setminus \{a_r\}$ and require $s \equiv 1 \pmod{2}$. CRT gives a reduced class b modulo $2 \prod_r r$; Dirichlet gives a prime s in it. If $s < Q$ then s is a covered odd prime, so $s \equiv a_r \pmod{r}$ for some r , contradicting $s \equiv b_r$. Hence $s \geq Q$. Now require $t \equiv a_r - s \pmod{r}$ for all r and $t \equiv 1 \pmod{2}$; every $a_r - s$ is nonzero mod r , so this is again a reduced class, and Dirichlet gives infinitely many primes t . Keep those with $s + t > Q + \max_r r$ and set $N = s + t$: it is even, satisfies the congruences, and has the explicit partition $N = s + t$. Proposition 2.9 excludes every prime below Q . $\square$

Remark 2.13 (Quantitative form for initial-segment covers). When the covering has one modulus for every prime $p \leq x$, the two residue classes above are reduced modulo $M = 2 \prod_{p \leq x} p$, and Linnik's theorem [8] supplies $s, t \ll M^L$. Hence $\log N \ll \log M = \theta(x) \ll x$. This bound is asserted only for initial-segment covers, not for arbitrary sparse finite ones.

2.4 Transfer from large prime gaps

Proposition 2.14. *If $[N - Q, N - 2]$ contains no prime, then no prime $q \leq Q$ is a Goldbach summand of N ; hence if N is Goldbach-representable, $\mathfrak{g}(N) > Q$.*

Proof. Every prime $q \in [2, Q]$ has $N - q \in [N - Q, N - 2]$, hence composite. $\square$

So a prime gap of length $\geq Q$ just below N forces $\mathfrak{g}(N) > Q$: prime gaps transfer to least Goldbach summands. The transfer is strictly one-directional, because forcing $\mathfrak{g}(N) > Q$ requires $N - q$ composite only for *prime* offsets, a far sparser target than a whole interval.

The asymmetry is quantitative, and we can measure it on our own records. For the 2692-digit desert of Table 1.1, the certified value is $\mathfrak{g} = 1\,134\,871$, which exceeds the largest certified prime gap known (1 113 106) while the integer realising it is about 6.9 times shorter in digits. Conversely, for the 237-digit height construction of the earlier draft of this work, the interval $[N - 1926, N - 2]$ is prime-free while $N - 1927$ passes BPSW: ordinary gap information gives only $\mathfrak{g}(N) > 1926$, whereas the certified value is 109 621. The construction gains two orders of magnitude precisely by ignoring composite offsets. Deserts are a relaxation of gaps, and the relaxation is worth roughly a factor of 7 in the size of the integer and two orders of magnitude in what a given integer certifies.

Writing $G(X)$ for the largest gap between consecutive primes below X , Ford–Green–Konyagin–Tao and Maynard [11, 12], sharpened in [13], give

$$G(X) \gg \frac{\log X \log_2 X \log_4 X}{\log_3 X},$$

improving Rankin [9]. Feeding the covering that underlies these bounds into Theorem 2.12 makes the transfer unconditional.

Theorem 2.15. *There are infinitely many even integers N , each with an explicit Goldbach representation, such that*

$$\mathfrak{g}(N) \gg \frac{\log N \log_2 N \log_4 N}{\log_3 N}.$$

Proof. The Erdős–Rankin method in the form of [13] supplies, for large x , residue classes $a_p \pmod{p}$ —one for each prime $p \leq x$ —whose union contains every integer of $[1, y]$ with $y = \lfloor cx \log x \log_3 x / \log_2 x \rfloor$.

Normalisation. If a_2 is odd, replace every a_p by $a_p - 1$; the shifted system covers $[0, y - 1]$. So we may assume $2 \mid a_2$, whence no odd integer of $[1, y - 1]$ is covered by the class modulo 2, and every odd prime $q < Q := y - 1$ satisfies $q \equiv a_r \pmod{r}$ for some *odd* prime $r \leq x$. Those pairs form a covering in the sense of Proposition 2.9; it covers far more than needed (all odd integers below Q), and self-covering ($r = q$, $a_r = 0$) is harmless.

Realisation. Theorem 2.12 yields infinitely many even N with an explicit partition and $\mathfrak{g}(N) \geq Q$, and by Remark 2.13 we may take $\log N \ll \theta(x) \ll x$.

Conclusion. Let C satisfy $\log N \leq Cx$. For large u the map $u \mapsto u \log u \log_3 u / \log_2 u$ is increasing, and each iterated logarithm of $\log N/C$ is $(1 + o(1))$ times the corresponding one of $\log N$; hence $\mathfrak{g}(N) \geq Q \gg \log N \log_2 N \log_4 N / \log_3 N$. Selecting one realisation for each large x gives infinitely many distinct values. $\square$

Together with the Granville–van de Lune–te Riele conjecture (1.2), this brackets the maximal order of $\mathfrak{g}$:

$$\underbrace{\frac{\log N \log_2 N \log_4 N}{\log_3 N}}_{\text{rigorous}} \ll \max_{N' \leq N} \mathfrak{g}(N') \lesssim \underbrace{(\log N)^2 \log \log N}_{\text{conjectural}}.$$

The gap between these is a genuine open problem, and nothing in this thesis closes it. What the remaining chapters do instead is settle a different and—we would argue—more tractable question: within the constructive paradigm that produces every known record, what is the exact cost of a desert, and is there any cheaper paradigm?

Chapter 3

The cost of a desert: a calibrated model and an engine

Chapter 2 reduced desert construction to a search: fix a partial cover, walk the progression $N = N_0 + kM$, and test the residual offsets. This chapter answers the obvious next question—how long does that take?—and then reports what happened when we ran it. The answer is a single exponent, and the exponent is accurate enough that most of the rest of this thesis can proceed by arithmetic rather than by experiment.

3.1 The failure exponent

Fix a partial cover with modulus $M = 2 \prod_{r \in \mathcal{R}} r$ and residual set U of prime offsets below Q . For a candidate $N = N_0 + kM$, the covered offsets are composite by construction; the candidate succeeds precisely when *every* residual complement $N - q$, $q \in U$, is also composite.

Two effects set the probability. First, each complement is a number of size about N , so by the prime number theorem it is prime with probability about $1/\log N$. Second—and this is the roughness bonus of Remark 2.8—the cover has already forced $N \equiv a_r$, so each complement $N - q$ avoids divisibility by every $r \in \mathcal{R}$ (for the offsets that were *not* killed by r), which raises its chance of being prime. Mertens’ theorem turns that into a multiplicative correction. Writing

$$\text{boost} = 2 \prod_{r \in \mathcal{R}} \frac{r}{r-1}, \quad (3.1)$$

the probability that one residual complement is prime is $\approx \text{boost}/\log N$, and, treating the $|U|$ residual complements as independent, the probability that a candidate survives all of them is

$$\left(1 - \frac{\text{boost}}{\log N}\right)^{|U|} \approx e^{-E}, \quad E = \frac{|U| \cdot \text{boost}}{\log N}.$$

Thus e^E candidates must be scanned per expected success. The independence step is a heuristic—it is the Hardy–Littlewood picture applied to a fixed arithmetic progression—and Chapter 6 shows that it is in a precise sense the *only* heuristic in play, because any correlation between two complements is forced to act through small shared divisors, which the cover already controls.

The formula explains the shape of the whole subject. Three levers move E :

- **More moduli** shrink $|U|$ but inflate boost and $\log M$. The balance between these is not obvious, and §4.2 shows it is an equilibrium rather than a budget constraint.

- **A bigger N** increases $\log N$ and so lowers E directly. This is why the height game is easy and the budget game is hard: digits are the cheapest resource in the problem.
- **A higher target Q** raises $|U|$ roughly in proportion to $\pi(Q)$, which is why record values of $\mathfrak{g}$ get expensive fast.

3.2 Calibration: does the model hold?

A cost model that has never been checked is a wish. We therefore instrumented every search to report, per slice of the progression, the measured pass rate $\hat{p}$ of the probable-prime tests, the mean number of surviving offsets per candidate, and the implied exponent

$$\hat{E} = \frac{(\text{mean survivors per candidate}) \times \hat{p}}{1},$$

which is the empirical analogue of E .

Across three independent covers and roughly 1.2×10^9 scanned candidates, we find

$$|\hat{E} - E| \leq 0.1 \text{ nats},$$

with per-slice values stable to within a couple of percent over hundreds of slices. Two consequences matter. The obvious one is that projections in later chapters can be made by arithmetic. The subtler one is a genuine bound on hidden structure: if some unexploited arithmetic correlation made constructed N systematically luckier than the model predicts, it would show up as a persistent gap between $\hat{E}$ and E . At 1.2×10^9 candidates, any such effect is worth at most 0.1 nats, a factor of 1.11. We use this as a quantitative constraint in Chapter 6, where it rules out an entire class of would-be mechanisms.

3.3 The engine

The search is embarrassingly parallel and utterly dominated by one primitive: a base-2 Fermat test on a number of the size of N . Everything else is bookkeeping designed to reduce the number of such tests.

Sieving. For each residual offset q and each small prime $s \nmid M$, the candidates k for which $s \mid N - q$ form an arithmetic progression in k , computable from one modular inverse. Marking those progressions removes most (k, q) pairs before any expensive test. The optimal sieve bound is empirical: at 199 digits we measure $B = 10^6$ to beat $B = 3 \cdot 10^6$ (901 versus 816 thousand candidates per second), because beyond that the sieve costs more than the tests it saves.

Amortisation across variants. A cover admits many *residue-swap variants*: reassigning some a_r produces a different progression with the same modulus M and the same $|U|$, hence the same E . These are free additional lottery tickets. Because they share M , they also share every modular inverse in the sieve, so caching that table across variants gave a measured +47% in throughput.

Survivor-ordered testing. Within a block, candidates differ in how many residual offsets survived the sieve. A candidate with a survivors succeeds with probability $(1 - p)^a$, so hits concentrate sharply in the low-survivor tail. Testing candidates in ascending order of a is result-preserving and measurably accelerates *discovery* (a factor of 1.83 in expected time-to-first-hit at record scale).

Selective testing. The natural next step is to *skip* the crowded candidates entirely. We implemented this with exact accounting: the engine computes the survivor histogram, weights each candidate by $(1 - \hat{p})^a$, and reports the fraction of total hit mass retained. This keeps the model honest— $\hat{p}$ remains unbiased, so calibration still works—while allowing a principled speed/coverage trade. At 199 digits, skipping the fullest 92% of each block took the raw scan rate from 1 772 to about 16 300 thousand candidates per second while retaining 27.6% of the hit mass: a net discovery rate about $3\times$ better.

The reason this works is worth stating, because it is not obvious. Each candidate dies at its *first* passing complement, so the expected number of Fermat tests per candidate is about $1/\hat{p} \approx 17$ regardless of how many survivors it has. Testing cost is therefore nearly flat in a , while success probability decays geometrically in a —so discarding high- a candidates is almost pure profit.

Hardware. The same algorithm on GPU is a different animal. A CUDA implementation using a bit-matrix sieve and Montgomery CIOS Fermat waves, validated bit-for-bit against the reference implementation, sustains about 2.5×10^6 tests/s on a T4 and 8.5×10^6 on an A100, versus roughly 6×10^3 on four CPU cores. That ratio, about 430, is $\log 430 \approx 6.1$ nats—and the E-gap between the rungs the two platforms can reach is likewise about 6. The record frontier is logarithmic in compute, and this is the cleanest illustration of it in our data: covers and strategy were already saturated at CPU scale, and the GPU rung was bought purely with throughput.

3.4 Search strategy: depth, breadth, and the ratchet

Given a fixed compute budget, how should it be spent? We ran the experiment both ways and the answer was unambiguous.

Breadth-first designs scan many residue-swap variants shallowly. Because the attainable N stays close to N_0 , every hit is small—good for the threshold game—but the design has a hard ceiling: when the variant pool is exhausted, the campaign ends with nothing.

Depth-first designs scan one progression far. The runway is unbounded, and because N grows only logarithmically with k , bad luck costs digits rather than terminating the campaign. Our 150-digit threshold record was found at $k = 6.21 \times 10^{10}$ of a full $[0, 7.5 \times 10^{10})$ scan—at 63.8% depth, a central draw against a cumulative expectation of 0.40.

The synthesis is what we call the **ratchet**: design a cover, run it until a hit, *bank a certified record*, then re-tune the cover for the next rung and repeat. At $Q \approx 10^5$ this produced the descending ladder $195 \rightarrow 179 \rightarrow 150$ digits, each rung a certified record in its own right. Under a fixed budget the ratchet dominates a single grand-slam campaign for a structural reason: it converts a lottery with one payout into a sequence of lotteries each of which pays, and each rung also re-calibrates $\hat{E}$ before more compute is committed to the next.

3.5 Results, including the ones that failed

Table 1.1 listed the successes. Honesty requires the rest of the distribution, so Table 3.1 reports the campaigns that produced no record. In each case we give the survival probability—the model’s own prediction of how likely it was to see nothing, given the compute actually spent—so the reader can judge whether the failure indicts the model or merely the dice.

Table 3.1: Campaigns that produced no record. “Survival” is $e^{-\lambda}$ where λ is the cumulative expected number of hits over the compute actually spent; a value near 1 means the campaign never had a real chance, a small value means it was unlucky.

Campaign	Target	Candidates	Survival	Outcome
E10 (budget game, $Q = 107\,720$)	$\mathfrak{g} > 107\,719$	$2.4 \cdot 10^8$	0.17	unlucky, in-model
Campaign 2 ($Q = 99\,991$, 168d cover)	$T(10^5)$ ladder	$5.6 \cdot 10^8$	0.22	unlucky, in-model
Campaign 3 ($Q = 122\,000$)	$\mathfrak{g} > 119\,419$	$4.0 \cdot 10^8$	0.93	stopped early

Two of these were genuinely unlucky: a 0.17 survival means the campaign had a 83% chance of producing a record and did not. That is a one-in-six outcome, not a broken model, and in both cases the per-slice statistics tracked the prediction throughout—which is precisely the discipline that lets us say so with a straight face rather than reaching for an explanation. The third was stopped by hand after 2.3% of its budget when we learned that a parallel effort was attacking the same target; its checkpoint is committed and it remains resumable.

The methodological point is that a calibrated model turns a negative result into information. Without E , three failed campaigns would be an embarrassment or an excuse. With E , they are three consistent measurements of the same exponent, and they are part of the 1.2×10^9 -candidate calibration that §3.2 rests on.

3.6 Certification

Every record in this thesis is packaged so that its verification is independent of its discovery. The pipeline is deliberately dull:

1. **Reconstruct.** From the committed cover and (N_0, M, k) , recompute N and check it against the stored decimal string.
2. **Exclude.** For every prime $q < \mathfrak{g}$, produce a witness: either the modulus r with $r \mid N - q$ (checked by division) or a strong base-2 compositeness witness (checked by modular exponentiation). Both are unconditional; a prime cannot fail either test.
3. **Represent.** Verify that $\mathfrak{g}$ is prime and that $N - \mathfrak{g}$ is prime, the latter by an Atkin–Morain ECPP certificate [16, 17] generated by PARI/GP [18].
4. **Re-verify independently.** A second checker, written against the record file rather than the search code, re-derives every witness and validates the certificate chain step by step—including the projective elliptic-curve arithmetic—without calling the library that produced it.
5. **Hash.** A SHA-256 manifest fixes the artifacts.

Step 4 is not ceremony. Re-verifying the two incumbent records at the start of this programme is what established the baseline the rest of the work is measured against, and the independent checker has since caught discrepancies in intermediate artifacts that the primary pipeline reported as clean. Verification is also far cheaper than discovery: it depends only on the final cover, the finite residual check, and the certificate—never on reproducing the heuristic path that found them.

Chapter 4

The frontier: where covering stops, and why

With a calibrated exponent in hand we can ask sharp questions. How small can a desert be made at a given target Q ? How large can $\mathfrak{g}$ be made under a given digit budget? And—the question this chapter is really about—is the answer determined by mathematics we could improve, or by a wall?

The answer turns out to be surprisingly clean, and surprisingly unlike what we expected when we started. The difficulty of the whole subject is an *integrality gap*: fractionally, the covering problem is trivial. All of the exponent, every last nat of it, is the price of rounding a fractional cover to an integral one.

4.1 The frontier, measured

Table 4.1 gives the achievable exponent as a function of the digit budget, at fixed target $Q = 10^5$: for each cap on the number of digits of N , we build the best cover we can and report $|U|$ and E .

Table 4.1: Cover frontier at $Q \approx 10^5$: the exponent achievable under a digit cap. Fewer digits force a smaller modulus, hence fewer moduli, hence a larger residual set *and* a smaller $\log N$ to divide by—the two effects compound, which is why the frontier steepens so sharply below 130 digits.

Digits of N	Moduli	$ U $	E
195	89	692	16.8
170	79	754	20.6
150	72	802	24.4
140	68	830	26.8
130	64	862	29.5
120	60	899	32.8
100	52	973	41.3

Table 4.2 does the complementary experiment: fix the digit budget at 199 and raise the target. This is the budget game’s own ladder, and it is the table that governs Chapter 7’s planning.

Table 4.2: The rung ladder at 199 digits: exponent and candidate count as the target Q rises, using the best cover we can build at each target (88 moduli, boost ≈ 11). Fleet-days assume the measured 7.9×10^5 candidates/s of a two-session GPU fleet.

Q	$ U $	E	e^E candidates	fleet time
122 000	867	20.83	$1.1 \cdot 10^9$	23 minutes
135 000	1 004	24.12	$3.0 \cdot 10^{10}$	11 hours
150 000	1 138	27.34	$7.5 \cdot 10^{11}$	11 days
175 000	1 335	32.07	$8.5 \cdot 10^{13}$	3.4 years
200 000	1 569	37.69	$2.4 \cdot 10^{16}$	950 years

The spacing is geometric—each rung costs 25 to 280 times the last—which has a pleasant strategic consequence exploited in Chapter 7: climbing *every* rung costs only about 4% more than jumping straight to the top, and banks a certified record at each step.

4.2 Cover size is an equilibrium, not a budget

A natural mental model is that the cover is limited by the digit budget: use as many moduli as $\log M \leq \log N$ allows. That model is wrong at record scale, and correcting it explains why the frontier is so flat.

Consider adding one more modulus $r = 461$ to the 88-modulus cover behind our budget-game work. It kills about 3.5 residual offsets, worth $-3.5 \times \text{boost}/\log N = -0.084$ nats. But it also multiplies boost by $461/460$, which raises the primality odds of every *surviving* residual complement, worth $+0.082$ nats. The net effect is $+0.00$ nats: the cover is at a stationary point.

Observation 4.1 (Boost–kills equilibrium). At $B = 10^{200}$ the cover behind our budget-game work spends $\log M \approx 437$ of the ≈ 461 available nats, so the digit budget is not tight; the cover stops growing because $dE = 0$, not because it has run out of room. Adding moduli past the equilibrium is neutral to slightly harmful.

This is why leftover digit budget buys nothing, why the frontier in Table 4.1 is smooth rather than kinked, and why—as we will see—only two things can actually move E : better rounding mathematics, or more compute.

4.3 The linear-programming relaxation is vacuous

The covering problem is a set-cover: choose one residue class per modulus so as to cover as many prime offsets as possible. Set-cover problems come with a standard relaxation—allow fractional choices—whose optimum lower-bounds the integral one and whose dual prices certify optimality. A natural research programme, proposed in our own earlier planning documents, was to solve that relaxation and publish the duals as a certificate of how close greedy is to optimal.

We solved it. The result kills the programme, and in doing so reframes the whole subject.

Proposition 4.2 (LP vacuousness). *Let $\mathcal{R}$ be the 88 moduli $3, 5, \dots, 457$ used at record scale and let the universe be the odd primes below Q . In the relaxation where each modulus r may distribute a unit of weight over its residue classes and a prime is covered to the extent that weight sits on its class, the optimum covers 100.00% of the primes—at both $Q = 122\,000$ and $Q = 200\,000$.*

Verification. Solved with HiGHS in about two seconds at each target. The reason is a mass count: modulus r covers a $1/(r-1)$ fraction of the odd primes in expectation, and $\sum_{r \in \mathcal{R}} 1/(r-1) \approx 1.86$. With 186% of the required covering mass available, a fractional assignment can spread it to leave nothing uncovered. $\square$

So the LP dual certifies nothing whatsoever about integral covers, and the §7.4-style “optimality certificate” programme is dead as posed. What replaces it is a much better statement:

Corollary 4.3 (The exponent is a rounding gap). *Every nat of the failure exponent is the cost of integrality. Concretely, at $Q = 122\,000$:*

$$\underbrace{15.6\%}_{\text{naive rounding}} \longrightarrow \underbrace{7.56\%}_{\text{adaptive greedy}} \longrightarrow \underbrace{0\%}_{\text{fractional optimum}}$$

uncovered primes, where the first figure is $e^{-1.86}$, the density left by independent rounding at the available mass. Each absolute percentage point of residual removed is worth about 2.7 nats—a factor of 15 in compute.

Greedy therefore sits almost exactly halfway, in halvings, between doing nothing clever and perfection. The rest of this chapter asks whether the remaining half can be taken.

4.4 Attacking the rounding gap

We fix a concrete benchmark: the 88 moduli above, the 11 474 odd primes below $Q = 122\,000$, and the greedy/coordinate-descent incumbent with $|U| = 867$. One prime saved is worth 0.0240 nats. The question is how much of the gap to 0 is reachable.

4.4.1 How much is there to win?

Before optimising, it is worth knowing the size of the prize. Sampling the space of assignments uniformly (choosing each a_r at random) gives $|U|$ with mean $\mu = 1377$ and standard deviation $\sigma = 21.0$. Two things are notable. First, σ is *below* the independence prediction of 34.8: the choices are negatively correlated, so the deep tail is thinner than a naive model suggests. Second, the design space has entropy $\ln \prod_r (r-1) = 435.7$ nats, so a first-moment (annealed) calculation puts the best assignment in the ensemble at roughly

$$u^* \approx \mu - \sqrt{2 \ln \Omega} \sigma \approx 758,$$

against greedy’s 867. Greedy already sits at $z = 24.3$ of the 29.5 standard deviations theoretically available. The honest prize is therefore about 2.6 nats—a factor of 14—and only if the annealed threshold is achievable, which for a clustered landscape it typically is not.

4.4.2 Exact local optimisation

We attacked the benchmark with large-neighbourhood search: freeze all but a window of 7–15 moduli and re-solve that window *exactly* with CP-SAT [30] over the primes left uncovered by the frozen part, then rotate windows. Windows were chosen three ways—by tightness (the moduli whose current class beats its best alternative by the smallest margin), randomly among tight

moduli, and uniformly—and the incumbent was enforced as a hard floor so the solver could only match or improve.

Across 41 windows, with time caps up to 300 seconds, the incumbent was matched every time and never beaten. To turn “not beaten” into “provably optimal at this scale” we added an LP certificate per window: solve the window’s own relaxation, and if $\lfloor \text{LP} \rfloor$ equals the incumbent’s coverage then no integral assignment of that window can do better. This certified 21 of 22 windows where it was computed. The single exception was a window of the smallest moduli—where, exactly as Proposition 4.2 predicts, the LP is loose.

We then went further and solved the *entire large-modulus endgame* at once: first the 30 largest moduli simultaneously, then the 44 largest—half the cover in a single exact subproblem. Both leave $|U| = 867$.

4.4.3 Global methods

Local exactness having failed, we tried methods that can cross basins.

Belief propagation with decimation. We wrote the assignment problem as a factor graph—one variable per modulus with domain $\mathbb{Z}/r$, one factor per prime contributing weight $e^{-\beta}$ when uncovered—and ran sum-product message passing, then repeatedly froze the most polarised modulus and re-ran (the standard decimation scheme [29]). Validation first: on a small instance where exhaustive enumeration is possible, BP tracks the exact $E[|U|]$ to 0.7% in the soft regime, and BP-guided decimation finds the exact optimum. On the real instance it returns 874 at $\beta = 1.5$ (and worse at higher β , where early freezes over-commit).

That is seven primes short of greedy, and it is the second-best construction we know for this instance—but it is not an improvement.

Basin structure. The most informative experiment was the cheapest. Take a solution produced by random-start coordinate ascent, which lands at $|U| \approx 968$, and apply the same exact-window machinery to it: twenty windows recover *none* of its 101-prime gap to greedy. Independent ascent solutions agree with each other on only 2.4% of their class choices—chance level—so there is no backbone to exploit, and crossover-style population methods have nothing to graft.

Observation 4.4 (The landscape). The assignment space is a field of mutually distant, locally-exact-stable basins spanning at least $|U| \in [867, 980]$. Exact local search cannot move between them at all; what distinguishes greedy is not the neighbourhood it explores but the *sequentially adaptive* way it constructs a solution, which lands it about 100 primes deeper than a typical basin.

4.4.4 Verdict, and one methodological caution

The greedy plateau survived every method we brought: 41 exact windows (mostly LP-certified), a half-cover exact block, BP decimation, and basin probing. We state the conclusion carefully, because the temptation to over-claim here is strong.

Observation 4.5 (Door-1 verdict). At $Q = 122\,000$ with these moduli, $|U| = 867$ is at or very near the quenched optimum. The annealed estimate of 758 is an *ensemble-existence* heuristic, not a constructive promise, and the clustering visible in Observation 4.4 is exactly the situation in which annealed and quenched optima diverge. The realistic prize is therefore 0 to 2.6 nats, probably nearer the bottom of that range.

The caution concerns mean-field methods. Naive belief propagation, run to low temperature, predicts $E[|U|] \rightarrow 35$ at $\beta = 4$ —heading for the fractional fantasy of Proposition 4.2 rather than

the achievable 867. The replica-symmetric approximation inherits precisely the LP's blindness to integrality. Anyone using message passing to estimate a threshold here needs the one-step replica-symmetry-breaking treatment with complexity bookkeeping; we flag this as the honest remaining theory item rather than quoting a number we do not trust.

4.5 What the frontier costs in money

Putting Table 4.1 together with measured throughput gives the price list in Table 4.3. It is worth stating in these terms because the strategic question in Chapter 7 is precisely which of these rungs is worth buying.

Table 4.3: Cost of the threshold game at $Q = 10^5$ by digit target, at 8.5×10^6 tests/s per A100 and cloud pricing current at the time of writing.

Digits	E	A100-years	approx. cost
140	28.1	0.10	\$1 000
135	29.7	0.49	\$5 000
130	31.2	2.3	\$24 000
125	32.9	12	\$125 000
120	34.7	73	\$770 000
100	44.1	$8.9 \cdot 10^5$	\$9.4 billion

The table makes the shape of the problem vivid. The frontier is fundable down to about 130 digits and institutional down to 120; sub-100 is seven orders of magnitude beyond plausible compute. Whether that last wall can be attacked by mathematics rather than money is the subject of the next chapter.

Chapter 5

Sub-100 digits: a reduction to two hard problems

Chapter 4 priced the threshold game at $Q = 10^5$ and found the 100-digit rung seven orders of magnitude beyond plausible compute. This chapter asks whether that wall is real. The answer is that it is real *within the paradigm*, and that stepping outside the paradigm leads directly to two well-known hard problems—which is a more useful outcome than either a proof or a construction, because it says exactly what would have to be broken.

It is worth being clear at the outset that sub-100-digit deserts *exist* in abundance. With no cover at all, the failure exponent at $B = 10^{100}$ is $E_{\text{random}} \approx 110$, so the density of qualifying integers is e^{-110} and the expected count below 10^{100} is about $10^{100} \cdot e^{-110} \approx 10^{52}$. Nothing here is an existence obstruction. The entire difficulty is constructive.

5.1 Two regimes

Write $A = \log M$ for the cover mass in nats and $\log B$ for the digit budget ($\log B \approx 230$ at 100 digits). Covers split into two regimes according to whether their modulus fits under the budget.

Regime I: $A \leq \log B$. The CRT progression itself supplies $e^{\log B - A}$ candidates, and the search costs e^E . Optimising at $Q = 10^5$, $B = 10^{100}$ gives a best cover of 44 moduli with $M \approx 10^{79}$, $|U| = 1060$, boost = 9.6, hence

$$E = 44.1, \quad e^E \approx 1.4 \times 10^{19} \text{ candidates.}$$

At the record engine's measured throughput of about 5×10^5 candidates per second this is roughly 9×10^5 GPU-years.

Remark 5.1 (A units error, corrected). An earlier version of this calculation reported 5×10^4 GPU-years, an error of a factor of 18: it divided the candidate count by the *test* rate (8.5×10^6 Fermat tests/s) rather than the *candidate* rate, forgetting that a candidate costs about 17 tests. The error was caught by cross-checking against an independent study of the same frontier, which measured a realisable $E = 43.66$ at 100 digits—within 0.4 nats of the figure above, and irreconcilable with the faster cost. We record the correction here because the discipline that caught it (always cross-check a headline cost against an independently derived one) is worth more than the number.

Regime II: $A > \log B$ (oversized covers). If the modulus is allowed to exceed the budget, coverage becomes excellent. All 9 592 primes below 10^5 can be covered by 355 moduli of total mass 1 019 digits; allowing each modulus to use $L = 16$ residue classes, 420 moduli leave only $|U| = 165$, giving

$$E = 10.2, \quad e^E \approx 27\,000 \text{ candidates.}$$

Twenty-seven thousand candidates is nothing. If we could produce such a design *together with* a representative below 10^{100} , sub-100 would fall this afternoon.

Solutions exist: the design space (which moduli, which classes) carries about 2 698 nats of entropy against $\log(M/B) = 2\,630$ nats of “distance to squeeze”, leaving a surplus of about +57 nats and hence roughly e^{57} valid (design, N) pairs below 10^{100} . But the *density* of designs whose representative lands below B is e^{-2630} . Enumeration is out by some 1 100 orders of magnitude.

Observation 5.2. Regime II is where sub-100 becomes easy *if and only if* small representatives can be found directly. That is the whole problem.

5.2 The small-representative problem is modular subset-sum

CRT reconstruction is additive. Writing the standard idempotent basis,

$$N \equiv \sum_{r \in S} a_r \cdot \frac{M}{r} \cdot \left(\left(\frac{M}{r} \right)^{-1} \bmod r \right) \pmod{M},$$

so “find a design whose representative is below B ” is exactly:

$$\text{minimise } \left| \sum_r c_r \bmod M \right| \text{ over choices } c_r \in C_r \text{ with } |C_r| = L, \text{ targeting the window } [0, B).$$

This is an inhomogeneous *modular subset-sum problem with per-position choices*. Its density—the ratio of available design entropy to the window squeeze—is

$$d = \frac{2\,698}{2\,630} \approx 1.026.$$

That is the worst possible value. Low-density instances ($d < 0.94$) fall to lattice reduction [21, 22]; very high density instances fall to birthday and k -tree methods [23]. Density ≈ 1 is precisely the regime where neither works and where the presumed hardness of subset-sum is concentrated. Concretely, our solution set has size about 2^{82} inside a design space of about 2^{3893} ; meet-in-the-middle gives 2^{1897} and Wagner’s k -tree with our list sizes still needs 2^{938} . Both are astronomically short.

5.3 The decoding shortcut, and why it fails

There is one route that would bypass the subset-sum problem entirely. Finding a small integer consistent with many congruences is a decoding problem for the Chinese Remainder code, and CRT codes have list-decoding algorithms [19, 20]. If the decoder’s radius were large enough, we could simply ask it for a small N agreeing with a chosen cover.

The relevant statement is Guruswami–Sahai–Sudan [19], Theorem 3: for a CRT code with moduli p_i , message bound K , and any non-negative integers ℓ and z_i , one finds in time $\text{poly}(n, \log N, \ell, \sum z_i)$ all codewords m with

$$\sum_i \alpha_i z_i \log p_i > \log(\ell + 1) + \frac{\ell}{2} \log \frac{K}{2} + \frac{1}{\ell + 1} \sum_i \binom{z_i + 1}{2} \log p_i,$$

where $\alpha_i = 1$ exactly when $m \equiv r_i \pmod{p_i}$. Optimising z and ℓ under uniform weights, with L allowed classes per position, gives an agreement-mass radius

$$A > \sqrt{L \cdot \log B \cdot P}, \quad P = \sum_{\text{pool}} \log p_i. \quad (5.1)$$

We verified (5.1) numerically against the exact finite- ℓ optimum: the ratio is 1.001–1.004 across pools from 5×10^3 to 10^5 and $L \in \{1, 2\}$.

Now the trap closes. The radius scales with the mass P of the *whole pool*, which must be far larger than the cover itself in order to supply the subset entropy that makes small representatives exist at all. Numerically, at pool bound 60 000 with $L = 16$: the pool mass is 59 816 nats, the decoding radius is 14 845 nats, and our cover mass is 2 860 nats. The decoder needs agreement $5.2\times$ larger than anything we can supply. Shrinking the pool lowers the radius but destroys the entropy that makes solutions exist.

Theorem 5.3 (Barrier; heuristic, first-moment). *For CRT-cover constructions of even $N < B$ with $\mathfrak{g}(N) > Q$, no parameter choice simultaneously satisfies*

- (a) $A < \log B + \log \binom{n}{s} + s \log L$ (*representatives exist*), and
- (b) $A > \sqrt{L \cdot \log B \cdot P}$ (*GSS-decodable*).

Hence sub-100 via oversized covers requires either list-recovery of CRT codes beyond the Johnson-type radius—an open problem in coding theory—or a subset-sum algorithm at density ≈ 1 .

Table 5.1: The barrier, evaluated. “Exists” is the largest cover mass for which representatives below B still exist; “decodes” is the GSS radius. The ratio never approaches 1.

pool bound	L	pool mass	A exists	A decodes	ratio
1 000	1	956	339	469	0.72
60 000	1	59 816	570	3 711	0.15
60 000	16	59 816	1 305	14 845	0.09
10^6	16	998 483	1 697	60 651	0.03

5.3.1 A retracted claim, and what it teaches

An earlier version of this analysis reported an *open window*: a range of pool sizes where, apparently, representatives existed and the decoder still worked. That claim was wrong and is withdrawn. The error is instructive enough to record.

The two conditions of Theorem 5.3 both depend on an agreement size s —how many positions the sought codeword matches. The retracted analysis computed the decoding radius at one value

of s and the existence bound at a *different* value (the one maximising existence, near $2n/3$), and compared the two numbers. Requiring both conditions at the *same* s closes the window at every pool size. The lesson generalises well beyond this problem: *a feasibility window assembled from two separately optimised bounds is worthless; couple the parameters first, then optimise once.*

We also had the barrier re-derived independently, by a route that shares no code with ours—a Legendre-transform treatment of the same optimisation—which reproduces the 14 845-nat radius and finds the gap closed at every pool size, with the best case still 28% short. Independent confirmation matters especially when the first version of a result was wrong.

5.4 Does lattice reduction beat the radius in practice?

Decoding radii are worst-case statements. Our instances are highly structured and, as computed above, admit roughly e^{57} solutions—so it is entirely reasonable to hope that lattice reduction succeeds far beyond the proved radius, as it often does elsewhere. We tested this directly rather than assuming either way.

We built a small CRT-decoding testbed: plant a known small solution in a random instance at a controlled agreement mass, then run LLL [24] on the standard Howgrave-Graham-style lattice and record whether the planted solution (or any other) is recovered. Sweeping the agreement mass across the Johnson-type radius A_J gives a sharp threshold:

- at $\ell = 30$, recovery succeeds above $1.028 A_J$ and fails below;
- at $\ell = 45$ the threshold tightens to about $1.02 A_J$, converging to the theoretical radius from above;
- below the threshold the failure is *total*: across 390 decodes we recovered zero non-planted roots, and no partial or near-solutions.

The last point is the one that matters. Solution abundance buys nothing: even with 10^7 alternative solutions available in the instance, the lattice returns nothing at all below the radius. The Johnson-type radius behaves here like a genuine algorithmic boundary rather than a proof artifact, which is what makes Theorem 5.3 a barrier and not merely a gap in our knowledge.

5.5 Where this leaves sub-100

1. **Regime I is the only currently viable route**, and it is out of reach: $E = 44.1$, about 9×10^5 GPU-years, seven orders of magnitude beyond plausible compute. The fundable frontier is 130–140 digits (Table 4.3).
2. **Two well-posed open problems now sit under the record**, either of which would collapse it: (i) list-recovery of CRT codes beyond the Johnson-type radius for structured, solution-abundant instances; (ii) modular subset-sum with per-position choices at density ≈ 1 where 2^{82} solutions exist. Both are of independent interest, which is the right way for a chapter like this to end.
3. **The one place the barrier does not apply** is a construction that certifies compositeness by something other than a congruence divisor. Every argument in this chapter accounts entropy in units of $\log r$ per excluded offset; a mechanism that escapes that accounting

escapes the barrier. Whether any such mechanism exists is the subject of Chapter 6, and the answer, within a precisely defined language, is no.

Chapter 6

A closure theorem for the covering paradigm

Every record in this thesis, and every record we know of in this problem, is built the same way: choose N so that each small prime offset q has $N - q$ divisible by some small prime fixed in advance. Chapter 5 showed that pushing this paradigm to sub-100 digits runs into two hard problems. The obvious next question is whether the paradigm itself can be escaped—whether some other mechanism could certify $N - q$ composite more cheaply than a congruence.

This chapter argues that it cannot, within a language precise enough to make the claim checkable. The result is not a theorem in the sense of a completed proof in a journal; it is a theorem in the sense of a stated claim, with a proof skeleton whose cases are individually established, two named conditionals, and an adversarial falsification campaign that failed to break it. We are explicit throughout about which is which.

6.1 The empirical precursor

Before formalising anything, we ran five bounded probes into candidate mechanisms, each with a predeclared quantitative gate. The metric is *compression*: offsets certified per nat of design entropy, benchmarked against covering (which manages roughly 4 400 offsets per nat at $r = 3$, falling to about 0.4 per nat at $r \approx 1\,200$).

- P1. Prime-power moduli.** Upgrading a modulus p to p^2 costs an extra $\log p$ nats and yields *zero* new kills: the class $q \equiv a \pmod{p^2}$ is a subset of the class mod p that is already killed (at $p = 7$ we count $227 \subset 1\,613$). Strictly dominated.
- P2. Polynomial value-set identities.** Take $N = m^k - c$, so that $N - q = m^k - d^k$ is algebraically composite whenever $q = d^k - c$. The best case is $k = 2$: $c = 398$ kills 113 offsets below 10^5 . But confining N to a k -th-power family surrenders $(1 - 1/k) \log B$ nats—115 nats at $B = 10^{100}$ —and the value set below Q has at most $O(Q^{1/2})$ members. The mechanism cannot scale past a few hundred offsets *at any price*.
- P3. Sierpiński–Riesel exponent families.** For $N_n = k \cdot 2^n + c$ the residue mod p cycles with period $\text{ord}_p(2)$. For any fixed candidate the induced constraint is one congruence class: ordinary covering, no entropy gain. (The exponent structure does correlate residues *across*

the family, which is a legitimate search-throughput trick—noted for the engine, not for the entropy budget.)

P4. Norm forms and splitting conditions. “ $N-q$ is a norm in K ” does not certify compositeness—rational primes with the right splitting behaviour are norms—and any usable splitting condition on q is a congruence condition modulo discriminant-related moduli. Covering in Galois-theoretic clothing.

P5. Residual bias. Could N be chosen so that residual complements are composite *more often* than the boost predicts? The banked campaigns answer directly: across three covers and 1.2×10^9 candidates, $|\hat{E} - E| \leq 0.1$. Any unexploited compositeness structure is worth at most 0.1 nats.

All five terminated at their gates. That is suggestive but not conclusive: five probes is five probes. The rest of this chapter turns the suggestion into a claim with a proof structure, a formal language, and an adversary.

6.2 The formal setting

Definition 6.1 (Construction scheme). Fix a bound B and target Q . A *construction scheme* is a pair (F, C) :

- a *candidate family* F : a parameter set $\Theta \subseteq \mathbb{Z}^m$ with a poly-size description and an evaluation map $N : \Theta \rightarrow 2\mathbb{Z} \cap [1, B)$ computable in $\text{poly}(\log B)$ time. Its *design cost* is $\text{cost}(F) = \log B - \log |N(\Theta)|$, the nats of search freedom surrendered;
- a *certificate map* C : on input (θ, q) with $q < Q$ prime, it outputs $\perp$ or a divisor witness $D(\theta, q)$ with $1 < D < N(\theta) - q$ and $D \mid N(\theta) - q$, verifiable in $\text{poly}(\log B)$ time.

The certificate is *planted* if the divisibility holds as an algebraic identity over the family (checkable symbolically), and *extracted* if C discovers the divisor per instance. Write $K(\theta)$ for the certified offsets and $U(\theta)$ for the residual ones.

The *effective exponent* $E_{\text{eff}}(F, C)$ is the logarithm of the expected work, in candidates, to find a θ all of whose residual complements are composite. For the classical cover, $E_{\text{eff}} = E$ of (1.3), calibrated to ± 0.1 nats in §3.2.

Theorem 6.2 (Divisor-Paradigm Closure; target statement). *Let (F, C) be a construction scheme whose certificate map is planted and expressible in the schema language L of §6.4. Assume:*

(H1) *Hardy–Littlewood-type heuristics for shifted primes: conditional on all divisibility data, the primality indicators of distinct complements are independent with the Mertens-adjusted rates;*

(H2) *no compositeness test with constant soundness on random rough inputs runs in $o(\text{one modular exponentiation})$*

Then $E_{\text{eff}}(F, C) \geq E^(Q, B) - o(1)$, where E^* is the covering frontier. Moreover, extracted certificates on any window-dense family of targets are factoring-equivalent, and certificate-free detection is already performed at its known-optimal cost by the standard engine.*

Informally: within L , nothing beats covering; outside planted- L , a scheme must either factor or test, and both are at known cost floors.

6.3 The three lemmas

6.3.1 Lemma 2 first: correlations are covering in disguise

We take the lemmas out of order because the second is the one that is actually proved, and it does a surprising amount of work.

Proposition 6.3. *Let N be even and $q_1 \neq q_2$ odd primes below Q . If $d > 1$ divides both $N - q_1$ and $N - q_2$, then $d \mid q_1 - q_2$, and hence every prime factor of d is smaller than Q .*

Proof. Subtract: d divides $(N - q_1) - (N - q_2) = q_2 - q_1$, and $0 < |q_1 - q_2| < Q$. $\square$

Trivial to prove, and yet it settles an entire class of speculative mechanisms. Any scheme that hopes to make the residual complements *jointly* luckier—choosing N so that its leftovers conspire to be composite together—must act through a divisor shared by two complements, and Proposition 6.3 says every such divisor is supported on primes below Q . But divisibility by small primes is exactly what the cover fixes and the sieve harvests. There is nothing left to exploit.

Lemma 6.4 (Correlation exhaustion). *Let $\mathcal{F}_{<Q}$ be the information generated by the events $\{s \mid N - q\}$ for primes $s, q < Q$. Then every two-complement divisor event is $\mathcal{F}_{<Q}$ -measurable; conditional on $\mathcal{F}_{<Q}$ (and on the sieve's data up to its bound), (H1) makes the residual primality indicators independent, so the success probability factorises as $\prod_q (1 - p_q)$ and E_{eff} decomposes as design cost plus $\sum_q p_q$ —which is the covering objective. Empirically, $|\hat{E} - E| \leq 0.1$ over 1.2×10^9 candidates bounds any violation.*

6.3.2 Lemma 1: entropy accounting for planted certificates

Lemma 6.5 (Entropy accounting). *Let (F, C) be planted. For each mechanism define its certified sub-image density δ_{eff} : the density, within a generic window of width Q at height B , of integers for which the scheme can exhibit its planted witness using poly-range parameters. Then the amortised design cost per certified offset is at least $\log(1/\delta_{\text{eff}}) - O(\log \log B)$, and for every family in L this is at least the covering cost.*

The proof is by cases, and the cases are where the content is. We treat them in §6.5 after fixing the language, because the arguments are family-specific and quantitative rather than uniform.

6.3.3 Lemma 3: the detector floor, as an open problem

Hypothesis (H2) deserves to be stated on its own, because it is the one genuinely open ingredient and it is interesting independently of Goldbach deserts.

Conjecture 6.6 (H2, sub-modexp compositeness detection). *Let n be uniformly random among integers in $[X, 2X]$ with no prime factor below $B_0 = (\log X)^{O(1)}$. Does there exist a randomised algorithm deciding compositeness of n with soundness $\geq 2/3$ in time $o(M(\log X) \cdot \log X)$ —asymptotically cheaper than one modular exponentiation?*

Known art: trial division and batch-gcd methods [27] certify exactly the small-divisor mass and have soundness $o(1)$ on rough inputs; Jacobi symbols are equidistributed on rough composites; Fermat, Euler, Miller–Rabin and Frobenius tests all cost $\Theta(M \log X)$. We are aware of no sub-modexp result in either direction, and no lower bound in any realistic model.

The downstream consumer is quantified: a detector at cost c and recall ρ improves throughput by $1/(c + (1 - \rho))$, so $c = 0.1$, $\rho = 0.9$ is a $5\times$ speedup on every search in this thesis—and, since record difficulty is logarithmic in compute, converts directly into records. The same primitive dominates GIMPS and PrimeGrid [35], so a positive answer would be felt well beyond this problem.

6.4 The schema language L

To make Theorem 6.2 checkable we fix a concrete grammar of certificate mechanisms. Each schema compiles to three things: an enumerator of the offsets it certifies, a symbolic divisor witness, and a design-cost term.

$$\begin{array}{lcl}
 L ::= & \text{Congruence}(d, a) & N \equiv a \pmod{d} \\
 | & \text{PolyImage}(f \in \mathbb{Z}[m], \deg \leq 4) & N - q = A(m)B(m) \\
 | & \text{ExpFamily}(N = j \cdot a^n + c) & \text{per-offset order conditions} \\
 | & \text{MultiPoly}(A, B \in \mathbb{Z}[u_1..u_v], \deg \leq 2, v \leq 3) & \\
 | & \text{NormForm}(K, \text{disc} \leq 10^4) & N - q = \text{Norm}_K(\alpha) \\
 | & \text{Cyclotomic}(\Phi_n, n \leq 200) & \\
 | & \text{Compose}(L, L) & \text{depth} \leq 2
 \end{array}$$

Two exclusions are deliberate. Mechanisms whose certificate requires a primality decision to define membership are circular and excluded; and per-instance divisor discovery is *extraction*, handled by the second horn of the theorem rather than by L .

Two thresholds make “beats covering” precise. A schema must certify more than $\text{CEIL} = 3\sqrt{Q} \approx 1\,048$ distinct prime offsets—placing it above the $\sqrt{Q}$ ceiling that limits identity families—and it must do so at a marginal cost below covering’s. The right benchmark for the second condition is subtle and we got it wrong at first (§6.7); the correct one is covering’s *average*:

$$\frac{437 \text{ nats}}{10\,607 \text{ offsets}} = 0.041 \text{ nats per offset},$$

not its marginal endgame cost of about 1.01. Covering wins on average, not at the margin, because one modulus spends $\log r$ nats and certifies $\pi(Q)/(r - 1)$ offsets *at once*, whereas every known algebraic mechanism buys $O(1)$ offsets per unit of entropy. Together with the design-entropy budget $\log B \approx 461$, the two conditions imply that a bulk mechanism needs cost $< \log B / \text{CEIL} = 0.44$ nats per offset.

6.5 Closing the families

6.5.1 Univariate identity families: a ceiling, not a price

A degree- d form takes at most $Q^{1/d}$ values below Q , so it can certify at most that many offsets *regardless of budget*: 349 at $d = 2$, 50 at $d = 3$, 19 at $d = 4$, 7 at $d = 6$. Since $\text{CEIL} = 1\,048$, every PolyImage with $d \geq 2$ is closed at any price, and $d = 1$ is ordinary covering. Cyclotomic schemas fall inside the same bound because $\varphi(n) \geq 2$ for every $n \geq 3$. This is the P2 probe generalised, and it retires two families outright.

6.5.2 ExpFamily: supply, not price

ExpFamily is the strongest non-covering family in the grammar and needs a quantitative argument. For $N = ja^n + c$, a prime s certifies the offset q when $(q - c)j^{-1}$ lies in the subgroup $\langle a \rangle \bmod s$, and the resulting condition confines the exponent n to one class modulo $\text{ord}_s(a)$: a cost of $\log \text{ord}_s(a)$ nats *per offset*.

Two facts then close it. First, a structural floor: $s \mid a^d - 1$ forces $a^d > s$, so $\text{cost} = \log d > \log \log_a s$ —cheap certificates require small moduli, which is exactly where covering is cheapest. Exhaustively over all $s < 10^5$ the cheapest is $\log 5 = 1.609$ (at $a = 7$, $s = 2801$).

Second—and this is the binding constraint—cheap certificates are *scarce*. A modulus s serves only the offsets landing in its order- d subgroup, a fraction d/s of them, so it supplies about $\pi(Q)d/s$ offsets: $s = 2801$ with $d = 5$ supplies about twenty. Solving the resulting purchase problem exactly—sort every $s < 10^5$ by $\log \text{ord}$, buy offsets cheapest-first—gives, for every base $a \in \{2, 3, 5, 7\}$:

Observation 6.7 (ExpFamily is over budget). Reaching 1048 certified offsets costs 1151 nats, against a total design budget of $\log B \approx 461$ nats: $2.5\times$ over. The most that any budget-feasible ExpFamily design can certify is 419 offsets.

6.5.3 MultiPoly: clusters, isolation, and orbits

Multivariate identities are the interesting case, and the one where we initially expected the theorem to break. Products of two binary quadratic forms have *window-dense* value sets—about 9 300 of the 17 983 prime offsets below $Q = 200\,000$ are representable—which blows straight past the univariate $\sqrt{Q}$ ceiling. If those representations could be exhibited, the family would be a genuine threat.

The cluster bound. Let $G = A \cdot B$ be homogeneous of total degree d in v variables, with value-set density exponent $\alpha = \min(1, v/d)$ (so the number of distinct values below X is about X^α). A first-moment count of k -element clusters of G -values inside a width- Q window below X gives $X^\alpha(X^{\alpha-1}Q)^{k-1}$, hence clusters exist only up to

$$k_{\max}(\alpha) = 1 + \left\lfloor \frac{\alpha \log X}{(1 - \alpha) \log X - \log Q} \right\rfloor,$$

finite whenever $\alpha < 1 - \log Q / \log X$. At $X = 10^{200}$, $Q = 2 \times 10^5$ that threshold is $\alpha^* = 0.9735$, and every combination with $v < d$ has $\alpha \leq (d - 1)/d \leq 0.9$:

Table 6.1: Cluster ceilings. A scheme needs about 18 000 certified offsets per candidate; every shared-variable identity family with $v < d$ delivers $O(1)$.

(d_1, d_2, v)	α	$k_{\max}$	(d_1, d_2, v)	α	$k_{\max}$
$(2, 2, 2)$	0.500	2	$(2, 2, 3)$	0.750	4
$(2, 3, 3)$	0.600	2	$(2, 3, 4)$	0.800	5
$(2, 4, 4), (3, 3, 4)$	0.667	3	$(2, 4, 5), (3, 3, 5)$	0.833	6
$(3, 4, 5)$	0.714	3	any $v < d$	≤ 0.9	≤ 13

We checked the density exponent empirically for the one in-grammar case with $v \geq 3$: for $(x^2 + y^2 + z^2)(x^2 + 2y^2 + 3z^2)$, counting distinct values in width- 10^4 windows at heights $10^6, 10^8, 10^{10}$ gives $\hat{\alpha} = 0.761$ against the predicted 0.750.

Above the threshold: isolation. For $v \geq d$ the value set is window-dense and existence no longer obstructs, so the closure must come from *access*. Around any known solution u_0 , the lattice points δ keeping $G(u_0 + \delta)$ inside a width- $2Q$ window form an ellipsoidal slab with principal extents $QX^{-(d-1)/d}$ along the gradient and $Q^{1/2}X^{-(d-2)/2d}$ along the Hessian directions. At $X = 10^{200}$, $Q = 2 \times 10^5$, $d = 4$ these are 10^{-145} and 10^{-47} : every extent is far below 1, so the only lattice point is $\delta = 0$. Window solutions are pairwise isolated; dense existence is non-constructive for geometric reasons, not complexity-theoretic ones.

Symmetry cannot help. One might hope an infinite symmetry group Γ moves between solutions. Three exhaustive cases: if Γ preserves G , navigation conserves the value—one offset per orbit. If Γ preserves one factor, it conserves that factor’s value, so the certificate always exhibits the same divisor; if that divisor is small enough to enumerate, the scheme is sieve-equivalent, and if it is balanced ($\approx X^{1/2}$), we are back to the isolated sliver. If Γ mixes generators from both factors, the reachable $(\log A, \log B)$ pairs form a coarse lattice with about $(\log X)^2 \approx 2 \times 10^5$ points against a target window of relative width 10^{-195} . Any remaining route must solve $G(u) = m$ for *prescribed* m —but a product-form representation of m is a nontrivial factorisation of m , which is the theorem’s extraction horn, not a new mechanism.

6.5.4 Compose: empty, not expensive

Finally, depth-2 composition. Our scorer treats a composition’s parts as independent and *sums* their certified offsets—generous—and Compose still peaked well below threshold. But the generosity conceals a stronger fact: two nontrivial family constraints on the same N are contradictory at record scale. Requiring $N = m^2 - c_1$ and $N = m'^2 - c_2$ forces $(m - m')(m + m') = c_1 - c_2$, so $m \leq |c_1 - c_2|$, whereas $N \sim 10^{200}$ needs $m \sim 10^{100}$. For ordinary shifts the solution sets are tiny— $c = 398, 4686$ admits five solutions, the largest with $m = 1073$. The intersection is empty at scale, so the only viable composition is family-plus-congruence: covering, plus one family capped at 349.

6.6 The falsification engine

A closure claim invites refutation, so we built the refuter ourselves. The engine compiles a schema to (enumerator, witness, cost), evaluates it against the live cover context, and applies the filter: a schema *passes* if it certifies more than CEIL distinct offsets at below-benchmark cost, with the design budget enforced.

Tier 0: calibration. Before trusting the engine we required it to reproduce the hand-derived numbers of §6.1 from primitives. It does: the best $m^2 - c$ family below $Q = 10^5$ is $c = 398$ with 113 killable offsets, matching P2 exactly, and the apparent discrepancy in the nats-per-residual-kill figure resolved as a $B = 10^{100}$ versus $B = 10^{200}$ context difference. Calibration also *corrected* the record: covering’s endgame is about 1.0 nats per offset at $Q = 122\,000$, not the 2.5 quoted earlier from a different target.

Tier 1: exhaustive. We swept 13661 schema instances across all seven families. **Zero passes.** The first run reported sixteen, which turned out to be a filter-semantics leak—congruence-containing schemas being scored against the escape criterion, when the covering

paradigm cannot by definition escape itself. After hardening, the near-miss board is topped by ExpFamily at 3.1–4.2 nats per residual kill, about $3\times$ covering’s endgame, exactly as P3 predicted.

Tier 2: generative. Finally we handed the problem to an adversary: an LLM-driven evolutionary search [33] whose fitness function was our scorer. The contract was designed to be unfoolable: candidate programs may *only* emit schema JSON, they run in a locked-down subprocess with no network, grading happens in a fresh interpreter with the trusted scorer, and a pass requires hand audit before it counts. A hostile candidate that simply prints `fitness=1.0` verifiably scores 0.

Forty-one evolved candidates produced zero passes and explored only four families, converging on the same ExpFamily optimum the exhaustive tier had already flagged as the closest near-miss and then stalling about $53\times$ short of the bar.

6.7 The referee broke before the theorem did

The most instructive thing that happened in this chapter was a failure of ours, not of the theorem.

Within about ten candidates the evolutionary search drove its reported fitness from 0.27 to 0.46, entirely inside ExpFamily, by pushing the mean per-offset cost from 3.71 down to 2.20 nats. It looked like a genuine climb toward falsification. It was not: the search had found an accounting hole in our scorer. ExpFamily was credited with about 10 416 certifiable offsets while being charged log ord nats *per offset* with the bill levied only against the 867 residual primes—so a design was credited with bulk coverage costing $10\,416 \times 2.20 \approx 22\,900$ nats of design entropy against a total budget of 461. A $50\times$ overdraft.

The audit protocol caught it before any claim was made, and the fix *strengthened* the mathematics. Charging per-offset mechanisms for every offset they claim, and capping coverage at the affordable fraction $\log B/\text{cost}$, produced three things we did not have before: the design-entropy budget constraint; the realisation that the honest benchmark is covering’s average 0.041 nats per offset rather than its marginal 1.01; and the supply-side purchase argument of Observation 6.7, which closes ExpFamily quantitatively rather than empirically.

Remark 6.8 (Methodological finding). Adversarial search against a formal claim is most valuable as a *referee-hardening* device, and should be reported that way rather than as a failed refutation. The properties that made this work are worth restating, because they are cheap and general: candidates emit data, never scores; grading runs in a separate trusted process; every quantity is re-derived from primitives; and a “pass” is a hand-audited event rather than an automatic conclusion. Under those conditions an adversary optimising against your evaluator is a free audit of your evaluator.

6.8 Verdict

What remains genuinely conjectural, stated plainly:

1. **(H1)**, Hardy–Littlewood independence—standard but unproved, supported here by the 1.2×10^9 -candidate calibration.

Table 6.2: Per-family closure status. 13 661 enumerated plus 41 evolved schema instances produced zero passes; each family closes for its own reason.

Family	Closure	Kind
Congruence	is the covering baseline; composite d dominated (P1)	proved
PolyImage ($k \geq 2$)	value set $\leq Q^{1/k}$, i.e. ≤ 349 offsets	proved, price-free
Cyclotomic	$\varphi(n) \geq 2$, same $\sqrt{Q}$ ceiling	proved, price-free
ExpFamily	purchase plan 1 151 nats vs 461; cap 419	proved, quantitative
MultiPoly ($v < d$)	cluster bound $k_{\max} \leq 13$	first moment
MultiPoly ($v \geq d$)	isolation + orbit conservation; else extraction	draft rigour
NormForm	reduces to MultiPoly	proved
Compose (depth 2)	intersection empty at 10^{200}	proved

2. **(H2)**, the sub-modexp detection question of Conjecture 6.6. A positive answer breaks the *cost floor*, not the covering monopoly.
3. The $o(1)$ terms in Lemma 6.5, and the value-equidistribution inputs to the cluster counting.
4. **Mechanisms outside L .** The language is concrete and deliberately finite. A mechanism that certifies compositeness by something that is neither a planted divisor identity nor per-instance divisor discovery is not excluded by anything here. We know of none, and the two structural horns—planted implies entropy accounting, extracted implies factoring—suggest why, but this is the honest frontier of the result.

The programme-level consequence is worth stating because it changes what should be worked on. If covering is not merely the best known paradigm but provably the only competitive one within L , then records are governed entirely by the two levers of Chapter 4: the rounding gap, and throughput. That is the subject of the final chapter.

Chapter 7

Outlook: three doors and a costed path

The preceding chapters were written in the order the work was done, which means they read as a narrowing. That narrowing is the result. This chapter states it positively: after Chapter 6, everything that could possibly improve a Goldbach-desert record passes through one of exactly three doors, and each door has a maximal form, a price, and a verdict.

7.1 The closure argument

Why only three? Because the three ways to change e^E are to change the mathematics that sets E , to change the rate at which candidates are examined, or to leave the paradigm in which E is defined. Chapter 4 showed the first is a rounding gap, Chapter 3 measured the second, and Chapter 6 closed the third within a concrete language. Two auxiliary facts complete the argument, and both are worth stating because they eliminate whole families of otherwise-attractive ideas:

- *Divisor information is fully harvested.* Any congruence cleverness applied to the search parameter k merely re-labels sub-progressions that the sieve already scores. There is no free information on the k side.
- *Correlation is covering* (Proposition 6.3), and beyond divisors the P5 measurement caps all remaining structure at 0.1 nats. So “luck-shaping” schemes have no purchase.

7.2 Door 1: the rounding gap (mathematics)

The exponent is entirely an integrality gap (Corollary 4.3), so better rounding is worth records. Measured prize: greedy’s $|U| = 867$ against an annealed ensemble threshold near 758, i.e. 0 to 2.6 nats—a factor of up to 14, compounding into *every* future rung.

Verdict from Chapter 4: resistant. Exact windows (LP-certified), half-cover exact blocks, and BP decimation all leave 867 untouched, and the landscape is a field of isolated basins in which the annealed threshold is probably not achievable. What has *not* been tried is a different *ensemble* rather than a better search of the same one. Specifically:

1. **Banded construction with an exact endgame.** Solve the small moduli exactly, choose the middle band with second-moment control on pairwise overlaps, and finish the large moduli as an exact weighted matching against the currently-uncovered set. This imitates *why* greedy wins—sequential adaptivity—while removing its myopia at the point where overlap waste is greatest.
2. **Import Erdős–Rankin technology constructively.** The FGKMT machinery [13] is, at bottom, a method for rounding a fractional cover of an interval with quantified loss; we used it in Theorem 2.15 for a lower bound but never as a construction. Its loss is $o(1)$ in an asymptotic regime that is not ours (their moduli reach $y/2$; ours stop at 457), so the question “what residual fraction is achievable at $\pi(Q) \approx 10^4$ with a budget of $\theta(457)$?” is genuinely open and thesis-sized.
3. **One-step replica-symmetry-breaking.** Needed to get an honest quenched threshold; naive belief propagation demonstrably slides toward the fractional fantasy (§4.4).
4. **Post the instance as a public optimisation challenge.** The benchmark is small, self-contained, and machine-checkable. Problems of exactly this shape attract strong heuristic solvers when posed publicly.

7.3 Door 2: throughput (compute)

Records are logarithmic in compute: each $10\times$ buys 2.3 nats, which is about four digits on the threshold game. Door 2 is therefore unglamorous and completely reliable. Three forms, priced.

Engineering already in hand. Porting the selective-testing scheme of §3.3 to the GPU fleet is worth $2\text{--}3\times$; batch remainder-tree sieving [27] to depth $10^9\text{--}10^{10}$ is worth perhaps $1.5\times$; replacing the fixed skip fraction with the exact marginal stopping rule adds 5–15%. None of this is research.

Volunteer computing. The search is a progress-free Poisson lottery with millisecond verification and no shared state—structurally the best-shaped volunteer workload that exists, and the exact shape on which PrimeGrid [35] and its relatives hold essentially every record in the neighbouring Sierpiński–Riesel problems, which are themselves covering problems. For calibration: a peak-era Folding@home-scale network is roughly 2×10^5 times our two-GPU fleet, which would put $Q = 200\,000$ (Table 4.2) about *two days* away; even 1% of that scale delivers it in months.

Silicon. The workload is 95% one instruction—a 660-bit Montgomery modular exponentiation with a fixed base. A crypto-mining-style ASIC plausibly buys $10\text{--}100\times$ an A100 per die at much better energy per test, so a single rack is worth $e^{5.5}$ to e^8 over today’s fleet; an FPGA farm is the no-NRE middle step. This is the capital-for-nats trade, and it is priced in the single-digit millions.

7.4 Door 3: the paradigm

Closed, within the language L (Chapter 6). The deliverable here has therefore flipped from *find a mechanism* to *finish the theorem and keep trying to break it*: complete the rigour of Lemma 6.5, settle (H2) or publicise it, and keep the falsification engine running as new schema families are proposed. If the closure claim is ever refuted, the payoff is enormous and immediate—but the sensible planning assumption is that it will not be.

7.5 Do we need a quantum computer?

Grover search [25] halves the exponent: e^E classical candidates become $e^{E/2}$ coherent oracle calls. Since $E = 37.7$ at $Q = 200\,000$ and 44.1 for sub-100, this looks decisive. It is not, and the arithmetic is worth doing once so that it need not be revisited.

One oracle call must evaluate the whole predicate reversibly: roughly $|U|$ modular exponentiations of 660-bit numbers with *no early exit*, since the classical trick of stopping at the first passing complement is unavailable in superposition. At fault-tolerant resource-estimate scales [26] this is on the order of 10^5 – 10^6 seconds per call. A classical candidate costs 1.25×10^{-6} seconds on our fleet. Writing C_q, W_q and C_c, W_c for the per-operation costs and the widths (numbers of machines), quantum wins only when

$$E > E^* = 2 \log \left[\frac{C_q}{W_q} \cdot \frac{W_c}{C_c} \right].$$

Table 7.1: The Grover crossover. Compare against $E = 37.7$ ($Q = 200\,000$) and $E = 44.1$ (sub-100).

Assumption	E^*
One fault-tolerant machine, $C_q = 10^6$ s	≈ 55
One thousand such machines	≈ 41
One thousand machines <i>and</i> a $1000\times$ cheaper oracle	≈ 27

With any non-miraculous constants the crossover sits at or above the hardest targets we care about—and at the crossover both approaches are equally infeasible. The honest answer is therefore: *no*. You would need thousands of much better quantum computers than anyone projects merely to *tie* with GPUs on this problem. Quantum walks for subset-sum face the same arithmetic in Chapter 5’s setting. We park the idea with numbers attached rather than dismissing it, so that it can be revisited if C_q ever approaches seconds.

7.6 Operation Staircase: a costed path to $g > 200\,000$

Putting the doors together gives a concrete plan. Its shape is dictated by Table 4.2: rungs are geometrically spaced, so climbing all of them costs about 4% more than jumping to the top, while banking a certified record and a fresh calibration at every step.

1. **Now.** $Q = 140\,009$ ($E \approx 25$) on the existing fleet: days. Bank it.

2. **Plus one week.** Land the engine work (selective testing on GPU, marginal stopping rule): $2\text{--}3\times$. Then $Q = 150\,000$ ($E = 27.3$): about four fleet-days. Bank it.
3. **Plus one month.** $Q \approx 165\,000$ ($E \approx 30$): two to three fleet-weeks with the upgraded engine. In parallel, Door 1's banded construction and the Erdős–Rankin import run as theory work; any ≥ 1 nat result compounds into every later rung.
4. **Plus one quarter.** $Q = 175\,000$ ($E = 32.1$): 1.4 fleet-years raw, so two to four months with the engine work and modest fleet growth. Decision gate: if Door 1 has delivered ≥ 2 nats, stay in-house; otherwise open Door 2 at scale.
5. **The wall.** $Q = 190\,000\text{--}200\,000$ ($E = 35.4\text{--}37.7$) needs about $e^{5.5}$ beyond step 4. Three interchangeable closers: a few hundred swarm or volunteer GPUs for a quarter; an ASIC/FPGA rack; or four nats of Door-1 mathematics plus a $10\times$ fleet. Any one suffices; any two make it comfortable.

Expected profile: records at 140k and 150k within weeks, 165k within a month or two, 175k inside a quarter, and 200 000 in two to four quarters—with the long pole attackable by money, community, or mathematics, whichever the programme can actually raise. Throughout, every rung is dual-verified by the pipeline of §3.6, and parallel workers claim disjoint variant sets in a committed ledger. That last point is not bureaucratic: one of our own campaigns was stopped mid-flight for duplicating another effort's target, which is a pure waste that coordination prevents.

7.7 Open problems

We close with the problems this thesis leaves in a state where someone else can pick them up. The first two would change the subject; the rest would advance it.

1. **Sub-modexp compositeness detection** (Conjecture 6.6). Is there a randomised test with constant soundness on rough inputs, cheaper than one modular exponentiation? Or a lower bound in any realistic model? A positive answer speeds up every prime-hunting project in existence; a negative one completes Lemma 6.5's third leg.
2. **The rounding gap for prime-residue covers.** Given $\pi(Q) \approx 10^4$ prime offsets and moduli up to 457, what residual fraction is achievable? Greedy gives 7.56%, the fractional optimum 0%, and the annealed ensemble threshold about 6.6%. Where is the quenched truth, and is it constructive?
3. **CRT list-recovery beyond the Johnson radius** for structured, solution-abundant instances (Chapter 5). Our experiments say the radius is a sharp algorithmic boundary in practice; a proof or an algorithm would both be significant.
4. **Modular subset-sum with per-position choices at density ≈ 1 ,** where 2^{82} solutions exist among 2^{3893} designs.
5. **Mechanisms outside L :** certify $N - q$ composite by something that is neither a planted divisor identity nor per-instance divisor discovery. This is the honest frontier of Chapter 6's claim.

6. **The smallest prescribed desert.** $S(q)$, the least even N with $\mathbf{g}(N) = q$ [4], packages into $T(Q) = \min_{q > Q} S(q)$; no construction here determines an exact minimum, and exhaustive verification stops at $4 \cdot 10^{18}$ while our constructions start at 10^{150} . The intervening 130 orders of magnitude are entirely unexplored.
7. **Closing the asymptotic sandwich.** Between $\log N \log_2 N \log_4 N / \log_3 N$ (rigorous, Theorem 2.15) and $(\log N)^2 \log \log N$ (conjectural (1.2)) there is a factor of about $\log N$. Does covering only prime offsets—rather than a whole interval—asymptotically beat the prime-gap scale?

7.8 A closing note on method

This thesis was produced with heavy delegation to automated agents: search campaigns, cover optimisation, literature checking, and an adversarial falsification engine. Three practices did more than any individual result to keep the output trustworthy, and we recommend them without reservation.

First, *calibrate before believing*. The failure exponent was checked against 1.2×10^9 candidates before it was used to make claims; the belief-propagation implementation was validated against exhaustive enumeration on a small instance before it was run on a large one. Both checks later caught real errors.

Second, *make negative results first-class*. Three campaigns produced no record. Reported with their survival probabilities they became measurements rather than embarrassments, and they now carry a substantial share of the model’s empirical support.

Third, *let the adversary audit the referee*. The single most useful event in Chapter 6 was an evolutionary search discovering a $50\times$ error in our own scorer. The audit protocol that caught it—candidates emit data, never scores; grading in a separate trusted process; every quantity re-derived from primitives; a “pass” is hand-audited—is cheap, general, and worth more than the theorem it was built to defend. We also retracted two claims in the course of this work (§5.3.1, Remark 5.1) and recorded both in place rather than quietly deleting them, which is the same discipline pointed at ourselves.

Appendix A

Data, code, and reproduction

Every claim in this thesis that rests on computation is backed by an artifact in the repository, and the artifacts are organised so that verification never requires reproducing discovery. This appendix is the index.

A.1 Certified records

Each record directory contains the canonical decimal string of N , the congruence cover, the reconstruction data (N_0, M, k) , per-offset exclusion evidence, an Atkin–Morain ECPP certificate for the prime complement, and a SHA-256 manifest.

Directory	Record	Verification
records/megagap_2692digit_g1134871/	2 692d, $g = 1\,134\,871$	witnesses + ECPP
records/threshold_150digit_g104527/	150d, $g = 104\,527$	witnesses + ECPP, re-verified
records/threshold_195digit_g100297/	195d, $g = 100\,297$	witnesses + ECPP
records/dual_197digit_g107719/	197d, $g = 107\,719$	witnesses + ECPP
records/budget_1e200_200digit_g112249/	200d, $g = 112\,249$	witnesses + ECPP
slop/goldbach/records/g1/	2 480d, $g = 1\,157\,341$	witnesses + ECPP
slop/goldbach/records/r200/	199d, $g = 119\,419$	witnesses + ECPP, re-verified
slop/goldbach/records/t100k/	ratchet 195 $\rightarrow$ 179d	witnesses + ECPP

Two independent verification paths exist: `verify_record.py` re-derives every per-offset witness from the record file alone, and `ecpp_verify.py` validates the elliptic-curve certificate chain step by step in projective coordinates without invoking the library that generated it.

A.2 Search engine and campaigns

`goldbach/search.py` The CRT-progression search engine: numpy presieve with cached modular inverses, survivor-ordered testing (`--sort-tests`), and selective testing with exact hit-mass accounting (`--skip-frac`, `--p-est`). Reports per-slice $\hat{p}$, survivors per candidate, $\hat{E}$, and retained hit-mass share—the telemetry behind §3.2.

`goldbach/cover.py`, `anneal.py`, `joint.py` Cover construction: greedy weighted set cover, simulated annealing, and the joint representative/coverage optimiser.

`goldbach/gpu/` The CUDA engine (bit-matrix sieve, Montgomery CIOS Fermat waves) and the multi-session fleet driver with git-committed resume marks.

`goldbach/phd/campaign{,2,3}/` The three campaigns of Table 3.1: specs, chunked drivers, and the committed `search.log` checkpoints. The logs are the primary evidence for the negative results, and they make each campaign resumable.

A.3 Frontier and optimisation studies

`goldbach/phd/wp8/pack_probe.py` Coordinate ascent with random restarts over class assignments; produces the rung ladder of Table 4.2 and the local-optimality finding.

`goldbach/phd/wp8/lp_bound.py` The set-cover LP relaxation (HiGHS) behind Proposition 4.2.

`goldbach/phd/wp9/door1/` The Door-1 benchmark and attacks: `instance_q122k.json` (sha256-pinned prime universe, the $|U| = 867$ incumbent), `d10_diagnostics.py` (landscape statistics), `d11_lns.py` (CP-SAT large-neighbourhood search with per-window LP certificates), `d11_basin_probe.py`, and `d12_bp.py` (belief propagation with decimation, including the exhaustive-enumeration validation).

A.4 Barrier and closure studies

`goldbach/phd/wp3_small_representative_memo.md` The sub-100 analysis of Chapter 5, including the retraction of §5.3.1 and the corrected costing of Remark 5.1.

`goldbach/phd/toy_crt/` The CRT-decoding testbed: planted instances, LLL decoding, and the threshold sweep across the Johnson-type radius.

`goldbach/phd/wp7_probes.md` The five mechanism probes of §6.1 with their predeclared gates.

`goldbach/phd/wp9/wp9_closure_theorem.md` The closure theorem in full working form, with proof-status markers throughout.

`goldbach/phd/wp9/engine/` The falsification engine: `tier0.py` (calibration against the hand-derived numbers), `exhaustive.py` (the 13 661-instance sweep) with `exhaustive_summary.json` (per-family aggregates and every near-miss; the full rows are regenerable and deliberately not committed), `scorer.py` (the trusted evaluator, with the budget accounting of §6.7), `harness.md` (the anti-reward-hack protocol), and `ae_runner.py` with `ae_seed_program.py` (the generative tier).

`goldbach/phd/wp9/h2_open_problem.md` Conjecture 6.6, stated self-contained for posting.

A.5 Reproducing the numerical claims

The claims of Chapters 4–6 are reproducible from the committed scripts, and all are cheap—seconds to minutes—because they are analyses rather than searches:

Claim	Command
LP vacuousness (Prop. 4.2)	<code>python3 phd/wp8/lp.bound.py 122000 200000</code>
Rung ladder (Table 4.2)	<code>python3 phd/wp8/pack_probe.py</code>
Door-1 diagnostics	<code>python3 phd/wp9/door1/d10_diagnostics.py</code>
Exact-window search	<code>python3 phd/wp9/door1/d11_lns.py 15 7 3 30</code>
BP validation + decimation	<code>python3 phd/wp9/door1/d12_bp.py</code>
Engine calibration (tier 0)	<code>python3 phd/wp9/engine/tier0.py</code>
Exhaustive schema sweep	<code>python3 phd/wp9/engine/exhaustive.py</code>
Single-schema score	<code>python3 phd/wp9/engine/scorer.py '{...}'</code>

Dependencies are `numpy`, `sympy`, `gmpy2`, `scipy` (HiGHS), and `ortools` (CP-SAT); the record verifiers require only Python 3, deliberately, so that verification has no dependency on the discovery stack.

A.6 A note on provenance

The programme ran as several parallel efforts sharing one repository, and the records in Table 1.1 were not all produced by the same effort. We have tried to be accurate about this throughout: where a record was produced elsewhere in the programme and re-verified here, the text says so, and the research log (`goldbach/phd/RESEARCH_LOG.md`) records the day-by-day provenance including the campaigns that failed, the two retractions, and the coordination failure that stopped Campaign 3. The theoretical contributions of Chapters 4–6—the LP vacuousness result, the Door-1 optimisation study, the sub-100 barrier analysis with its retraction, and the closure theorem with its falsification engine—were produced by the work reported here.

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